PHYSICS

1) The potential of electrostatic field is given by $\phi = xy$, where x & y are coordinate. Which of the following is correct :-

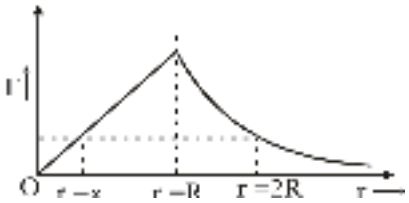
(1) Electric field at all points on positive x-axis is in positive y-direction.

(2) Electric field at all point of positive y-axis is in negative x-direction.

(3) Electric field at (2, 2) is $\sqrt{2}$ N/C

(4) Electric field at point (x, y) makes an angle $\theta = \tan^{-1}\left(\frac{y}{x}\right)$ with x-axis.

2) Variation in electric field for uniformly charged sphere is shown in figure then value of x :-



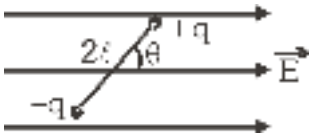
(1) $\frac{R}{3}$

(2) $\frac{R}{4}$

(3) $\frac{R}{2}$

(4) $\frac{3R}{8}$

3) The correct formula for total potential energy of the given system is $\left(k = \frac{1}{4\pi\epsilon_0}\right)$:



(1) U = –pE

(2) U = +pE

(3) U = +pEcos θ

(4) U = – pEcosθ – $\frac{kq^2}{2\ell}$

4) What is the amount of charge possessed by 1 g of electrons? Given that mass of an electron is 9.1×10^{-31} kg.

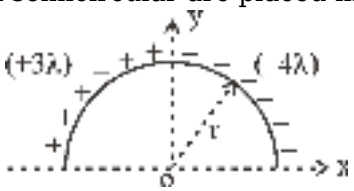
(1) 6.25×10^{10} C

(2) 1.76×10^{11} C

(3) 1.76×10^{8} C

(4) 1.25×10^{10} C

5) A semicircular arc placed in xy plane is charged as shown in figure. The electric field at centre 'O' is :-



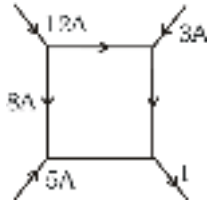
(1) $\frac{5\sqrt{2}\lambda}{4\pi\in_0 r}$, above +x direction

(2) $\frac{5\sqrt{2}\lambda}{4\pi\in_0 r}$, below +x direction

(3) $\frac{5\sqrt{2}\lambda}{4\pi\in_0 r}$, along + x direction

(4) $\frac{25\lambda}{4\pi\in_0 r}$, along + x direction

6) In fig. a network of currents is shown. The magnitude of currents is shown here. The current I will be :-



(1) –3A

(2) 3A

(3) 13A

(4) 20A

7) A galvanometer can be converted into an ammeter by connecting :-

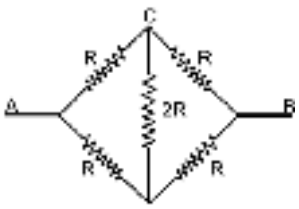
(1) Low resistance in series

(2) High resistance in parallel

(3) Low resistance in parallel

(4) High resistance in series

8) The equivalent resistance between A and B in the given circuit will be:



(1) R

(2) 2R

(3) $\frac{R}{2}$

(4) 6R

9) At what temperature will the resistance of a copper wire become three times its value at 0°C? [Temperature coefficient of resistance for copper = 4×10^{-3} per °C] :-

(1) 400°C

(2) 450°C

(3) 500°C

(4) 600°C

10) An air-filled parallel plate capacitor is to be constructed which can store 12μC of charge when operated at 1200V. What can be the minimum plate area of the capacitor ?
[Dielectric strength of air = 3×10^6 V/m]

(1) 0.38 cm^2

(2) 0.30 m^2

(3) 0.45 cm^2

(4) 0.45 m^2

11) A parallel plate capacitor has plates of unequal area. The larger plate is connected to the positive terminal of the battery and the smaller plate to its negative terminal. Let Q_+ and Q_- be the charges appearing on the positive and negative plates respectively :-

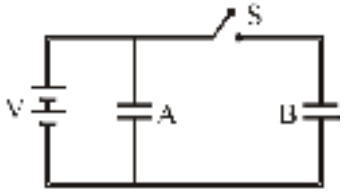
(1) $|Q_+| > |Q_-|$

(2) $|Q_+| = |Q_-|$

(3) $|Q_+| < |Q_-|$

(4) The information is not sufficient to decide the relation between Q_+ and Q_-

12) Figure given below shows two identical parallel plate capacitors connected to a battery with switch S closed. The switch is now opened and the free space between the plate of capacitors is filled with a dielectric of dielectric constant 3. What will be the ratio of total electrostatic energy stored in both capacitors before and after the introduction of the dielectric :-



(1) 3 : 1

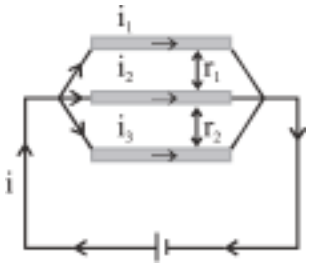
(2) 5 : 1

(3) 3 : 5

(4) 5 : 3

13) Three long straight wires are connected parallel to each other across a battery of negligible

internal resistance. The ratio of their resistances are 3 : 4 : 5. What is the ratio of distances of middle wire from the others if the net force experienced by it is zero :-



(1) 4 : 3

(2) 3 : 1

(3) 5 : 3

(4) 2 : 3

14) Two long identical conducting wires AOB and COD are placed at right angles to each other. The wire AOB carries a current I_1 and COD carries a current I_2. The magnetic field on a point lying at a distance d from O, in a direction perpendicular to the plane of the wires AOB and COD, will be given by-

(1) $\frac{\mu_0}{2\pi}\left(\frac{I_1 + I_2}{d}\right)^{1/2}$

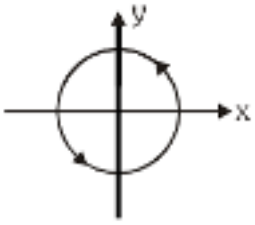
(2) $\frac{\mu_0}{2\pi d}\left(I_1^2 + I_2^2\right)^{1/2}$

(3) $\frac{\mu_0}{2\pi d}(I_1 + I_2)$

(4) $\frac{\mu_0}{2\pi d}\left(I_1^2 + I_2^2\right)$

15)

Current i = 2.5 A flows along the circle $x^2 + y^2 = 9\ cm^2$ (here x & y in cm) as shown



Magnetic field at point (0, 0, 4 cm) is

(1) $\left(36\pi \times 10^{-7}T\right)\hat{k}$

(2) $\left(36\pi \times 10^{-7}T\right)\left(-\hat{k}\right)$

(3) $\left(\frac{9\pi}{5} \times 10^{-7}T\right)\hat{k}$

(4) $\left(\frac{9\pi}{5} \times 10^{-7}T\right)\left(-\hat{k}\right)$

16) 4 gm of steam at 100°C is added to 20 gm of water at 46°C in a container of negligible mass. Assuming no heat is lost to surrounding, the mass of water in container at thermal equilibrium is. (Latent heat of vaporisation = 540 cal/gm. Specific heat of water = 1 cal/gm-°C).

(1) 18 gm

(2) 20 gm

(3) 22 gm

(4) 24 gm

17) A proton moving with a velocity 3×10^5 m/s enters in a magnetic field of 0.3 tesla at an angle of 30° with the field. The radius of curvature of its path will be :
(e/m for proton = 10^8 coulomb/kg)

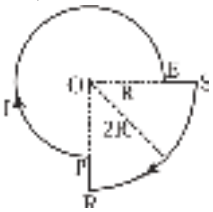
(1) 2 cm

(2) 0.5 cm

(3) 0.02 cm

(4) 1.25 cm

18) A current I flowing through the loop as shown in figure. The magnetic field at centre O is :-



(1) $\frac{7\mu_0 I}{16R}\otimes$

(2) $\frac{7\mu_0 I}{16R}\odot$

(3) $\frac{5\mu_0 I}{16R}\otimes$

(4) $\frac{5\mu_0 I}{16R}\odot$

19) A solid metal cube of edge length 2 cm is moving in the positive y-direction at a constant speed of 6 ms^{-1} .There is a uniform magnetic field of 0.1 T in the positive z-direction. The potential difference between the two faces of the cube perpendicular to the x-axis is

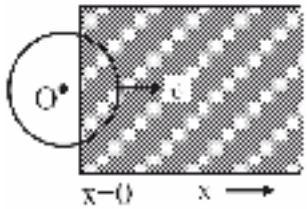
(1) 12 mV

(2) 1 mV

(3) 2 mV

(4) 6 mV

20) A constant magnetic field of 1 T is applied in the x > 0 region. A metallic circular ring of radius 1m is moving with a constant velocity of 1 m/s along the x-axis. At t = 0s, the centre of O of the ring is at x = – 1m. What will be the value of the induced emf in the ring at t = 1s? (Assume the velocity of the ring does not change.)



(1) 1 V

(2) 2π V

(3) 2 V

(4) 0 V

21) A coil of inductance 300 mH and resistance 2Ω is connected to a source of voltage 2V. The current reaches half of its maximum value in :

(1) 0.05 second

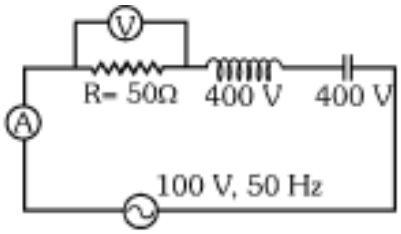
(2) 0.1 second

(3) 0.15 second

(4) 0.3 second

22)

In given LCR circuit, the voltage across the terminals of a resistance & current will be–



(1) 400V, 2A

(2) 800V, 2A

(3) 100V, 2A

(4) $100\sqrt{2}$V, 4A

23) A coil has a resistance 10 Ω and an inductance of 0.4 Henry. It is connected to an AC source of 6.5V, $\frac{30}{\pi}$ Hz. The average power consumed in the circuit, is :-

(1) $\frac{5}{8}$W

(2) $\frac{4}{3}$W

(3) $\frac{3}{8}$W

(4) $\frac{6}{7}$W

24) **Assertion :** Displacement current goes through the gap between the plates of a capacitor when the charge of the capacitor does not change.
Reason : The displacement current arises in the region in which the electric field is constant with time.

(1) If both assertion and reason are true and reason is the correct explanation of assertion.

(2) If both assertion and reason are true but reason is not the correct explanation of assertion.

(3) If assertion is true but reason is false.

(4) If both assertion and reason are false.

25) If radius of first Bohr orbit in hydrogen is r, then radius of second Bohr's orbit in hydrogen will be :-

(1) 2r

(2) 8r

(3) 4r

(4) $2\sqrt{2}\,r$

26) For an equilateral prism, it is observed that when a ray strikes grazingly at one face it emerges grazingly at the other. Its refractive index will be:-

(1) $\sqrt{3}$

(2) $\frac{2}{\sqrt{3}}$

(3) 2

(4) data not sufficient

27) **Statement-1 :** The formula connecting u, v and f for a spherical mirror is valid only for mirrors whose sizes are very small compared to their radii of curvature.
Statement-2 : Laws of reflection are strictly valid for plane surfaces, but not for large spherical surfaces.

(1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1

(3) Statement-1 is True, Statement-2 is False

(4) Statement-1 is False, Statement-2 is True

28) Two identical thin plano-convex glass lenses (refractive index 1.5) each having radius of curvature of 20 cm are placed with their convex surfaces in contact at the centre. The intervening space is filled with oil of refractive index 1.7. The focal length of the combination is :-

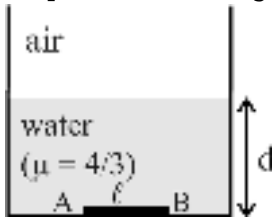
(1) –25 cm

(2) –50 cm

(3) 50 cm

(4) –20 cm

29) AB is an object dipped in water at a depth of d. Its length is □. It is seen from air at near normal incidence. The length of the image is :



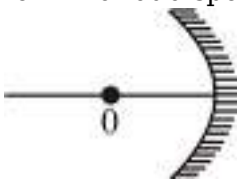
(1) □

(2) $\mu\ell$

(3) $\frac{\ell}{\mu}$

(4) none of these

30) An object 'O' is placed at a distance of 100 cm in front of a concave mirror of radius of curvature 200 cm as shown in the figure. The object starts moving towards the mirror at a speed 2 cm/s. The



position of the image from the mirror after 10s will be at ..... cm.

(1) 100

(2) 200

(3) 300

(4) 400

31) A plano convex lens of refractive index μ_1 and focal length f_1 is kept in contact with another plano concave lens of refractive index μ_2 and focal length f_2. If the radius of curvature of their spherical faces is R each and $f_1 = -2f_2$, then μ_1 and μ_2 are related as :

(1) $\mu_1 + \mu_2 = 3$

(2) $2\mu_1 - \mu_2 = 1$

(3) $2\mu_2 - \mu_1 = 1$

(4) $3\mu_2 - 2\mu_1 = 1$

32) A Young's double slit apparatus is shown in figure :-



	Column-I		Column-II
(A)	Source S move upwards	(P)	Central fringe moves upwards
(B)	A thin glass slab is placed in front of slit S_2	(Q)	Central fringe moves downward
(C)	Whole apparatus is immersed in water	(R)	Fringe width remains constant
(D)	Diameter of slit S_2 is doubled	(S)	Fringe width decreases
		(T)	Intensity of central maxima increases

(1) A→P, R ; B → Q, R ; C → R, T ; D → R, T

(2) A→Q, R ; B → Q, R ; C → S ; D → R, T

(3) A→P, R ; B → P, R ; C → S ; D → Q, R

(4) A→Q, R ; B → P, R ; C → R, T ; D → R, T

33) In case of diffraction pattern of a single slit under polychromatic illumination first minima of wavelength λ_1, is found to coincide with the second maxima of wavelength λ_2, then -

(1) $3\lambda_1 = 2\lambda_2$

(2) $2.5\lambda_1 = \lambda_2$

(3) $\lambda_1 = 2.5\lambda_2$

(4) $5\lambda_1 = \lambda_2$

34) A beam of natural light falls on a system of 5 polaroids, which are arranged in succession such that the pass axis of each polaroid is turned through 60° with respect to the preceding one. The fraction of the intensity of incident light that passes through the system is :

(1) $\frac{1}{64}$

(2) $\frac{1}{32}$

(3) $\frac{1}{256}$

(4) $\frac{1}{512}$

35) A nucleus of mass number 220, initially at rest, emits an alpha particle. If the Q value of the reaction is 12.25 MeV, The kinetic energy of the emitted α-particle will be :-

(1) 9.6 MeV

(2) 10.8 MeV

(3) 12.03 MeV

(4) 13.6 MeV

36) If the binding energies per nucleon of a deuteron and an alpha particle are 1.125 MeV and 7.2 MeV respectively, then the more stable of the two is :-

(1) Deutron

(2) Alpha-particle

(3) Both (1) and (2)

(4) Sometimes deutron and sometimes alpha particle

37) In a photocell bichromatic light of wavelength 2475 Å and 6000 Å are incident on cathode whose work function is 4.8 eV. If a uniform magnetic field of 3×10^{-5} Tesla exists parallel to the plate, the radius of the path described by the photoelectrons will be (mass of electron = 9×10^{-31} kg)

(1) 1 cm

(2) 5 cm

(3) 10 cm

(4) 25 cm

38) A proton & an electron both have energy 50eV
Assertion :- Both have different wavelength
Reason :- Wavelength depends on energy and not on mass.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.
(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
(3) **Assertion** is true but **Reason** is false.
(4) **Assertion** is false but **Reason** is true.

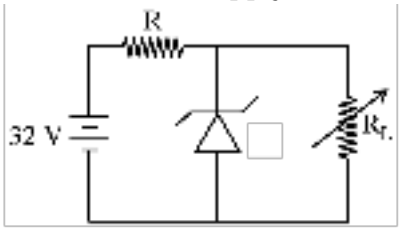
39) The de-broglie wavelength of an electron is the same as that of a photon. If velocity of electron is 25% of the velocity of light, then the ratio of kinetic energy of electron and energy of photon will be :-

(1) $\frac{1}{2}$
(2) $\frac{1}{8}$
(3) $\frac{8}{1}$
(4) $\frac{1}{4}$

40) **Assertion :** Net charge of p-type or n-type semiconductor is zero.
Reason : In p type semiconductor number of holes are greater than number of electrons.

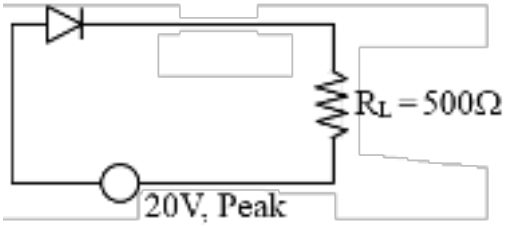
(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

41) A 24 V, 600 mW Zener diode is used to provide a 24 V stabilized supply to a variable load R_L as shown in the figure. The value of the resistance R is :-



(1) 320 Ω
(2) 640 Ω
(3) 960 Ω
(4) 1280 Ω

42) In circuit shown the barrier voltage of diode is 0.7 V and diode resistance is 10Ω. The match the

physical quantities :- 

Column-I		Column-II	
(A)	Peak current in diode (in mA)	(P)	37.8
(B)	Peak voltage(in volts) at the end of load	(Q)	40.0
(C)	Peak current (in mA) if diode is ideal	(R)	20.0
(D)	Peak voltage (in volts) at the ends of loads if diode is ideal	(S)	18.9

(1) A → Q ; B → R ; C → P ; D → S

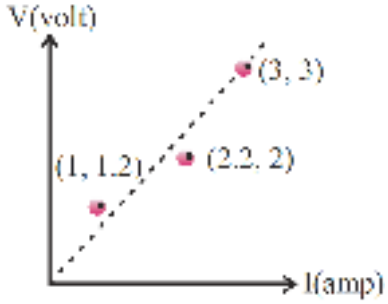
(2) A → P ; B → S ; C → R ; D → Q

(3) A → P ; B → S ; C → Q ; D → R

(4) A → S ; B → P ; C → Q ; D → R

43)

In the measurement of resistance of a wire using Ohm's law, the plot between V and I is drawn as shown. The resistance of the wire is -



(1) 0.833 Ω

(2) 0.95 Ω

(3) 1 Ω

(4) 1.1 Ω

44) To know the resistance G of a galvanometer by half deflection method, a battery of emf V_E and resistance R is used to deflect the galvanometer by angle θ. If a shunt of resistance S is needed to get half deflection then G, R and S are related by the equation :

(1) 2S (R + G) = RG

(2) S (R + G) = RG

(3) 2G = S

(4) 2S = G

45) A travelling microscope is focussed on a mark on a piece of paper and the scale reading is 5.25 cm. A rectangular block of glass is placed on paper, and the microscope is raised and focused again on the mark, then microscope reading is 11.25 cm. Now a layer of lycopodium powder is sprinkled on the block of glass and viewed the top surface. The reading obtained is 23.25 cm. The refractive index of material of block is :

(1) 1.32

(2) 1.45

(3) 1.5

(4) 1.64

CHEMISTRY

1) **Statement I** : Brown ring test can be done for NO_3^- in presence of NO_2^-.
Statement II : Oxidation state of iron is changing from +2 to +1 in the brown ring complex.

(1) Statement I is false but statement II is true.

(2) Both statement I and statement II are false.

(3) Statement I is true but statement II is false.

(4) Both statement I and statement II are true.

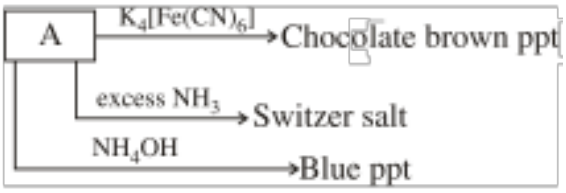
2) $K_2Cr_2O_7 + NaCl + H_2SO_4 \rightarrow A$ + other product
$A + NaOH \rightarrow B$ + other product
B + lead acetate $\xrightarrow{CH_3COOH}$ Yellow ppt (C)
A,B,C respectively :

(1) CrO_2Cl_2, Na_2CrO_4,$PbCrO_4$

(2) Na_2CrO_4, $PbCrO_4$, CrO_2Cl_2

(3) $PbCrO_4$, CrO_2Cl_2, Na_2CrO_4

(4) CrO_2Cl_2, $PbCrO_4$, Na_2CrO_4

3) 

Which ion is 'A'.

(1) Cu^{+2}

(2) Fe^{+3}

(3) Ca^{+2}

(4) Zn^{+2}

4) **Assertion :** Test of NH_4^+ cannot be done within group analysis.
Reason : During group analysis several times NH_4^+ compound is added at the different steps.

(1) Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

5) Identify the **incorrect** statement :

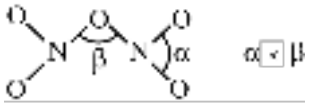
(1) Boron trioxide is acidic, aluminium and gallium trioxides are amphoteric while indium and thallium trioxides are basic.
(2) The hybridisation of Al in $[Al(H_2O)_6]^{3+}$ is d^2sp^3.
(3) Two isotopes of Boron, namely $_{10}B(19\%)$ and $_{11}B(81\%)$ are known.
(4) The relative stability of group 13 elements in +1 oxidation state varies as Al < Ga < In < Tl.

6) Match **List-I** with **List-II** :

	List-I (Oxoacids of Sulphur)		**List-II (Bonds)**
A.	Peroxodisul-phuric acid	I.	Two S-OH, Four S=O, One S-O-S
B.	Sulphuric acid	II.	Two S-OH, One S=O
C.	Pyrosulphuric acid	III.	Two S-OH, Four S=O, One S-O-O-S
D.	Sulphurous acid	IV.	Two S-OH, Two S=O

Choose the **correct** answer from the options given below:
(1) A-III, B-IV, C-I, D-II
(2) A-I, B-III, C-IV, D-II
(3) A-III, B-IV, C-II, D-I
(4) A-I, B-III, C-II, D-IV

7) Which of the following option is incorrect ?

(1) Melting point :- As > Sb > Bi > P > N
(2) Nitrogen has two stable isotopes ^{14}N & ^{15}N
(3) 

(4) $2Se_2Cl_2 \rightarrow SeCl_4 + 3Se$

8)

Which of the following will have three stereoisomeric forms :-
(i) $[Cr(NO_3)_3(NH_3)_3]$ (ii) $K_3[Co(C_2O_4)_3]$
(iii) $K_3[CoCl_2(C_2O_4)_2]$ (iv) $[CoBrCl(en)_2]^{+1}$

(1) iii, iv

(2) i, ii and iv

(3) Only iv

(4) All

9) In which of the following molecule orbital of different principle quantum number are used in the hybridization :-

(1) IF_7

(2) $[Ni(CO)_4]$

(3) $[PtCl_4]^{-2}$

(4) $K_4[Fe(F_6)]$

10) The IUPAC name of $K_2[Cr(CN)_2O_2(O)_2(NH_3)]$ is:-

(1) Potassiumamminedicyano dioxoperoxochromate(VI)

(2) Potassiumamminecyano peroxodioxo chromium (VI)

(3) Potassiumamminecyano peroxodioxo chromium (V)

(4) Potassiumamminecyano peroxodioxo chromate (IV)

11) For the complex, $[Fe^{II}(en)_3]_n\ [Co^{III}(NH_3)_2(NO_2)_4]_2$ the value of n is :-

(1) 1

(2) 2

(3) 3

(4) 4

12)

Which of the following order is not correct :-

(1) $[Fe(H_2O)_6]^{+3} < [Fe(CN)_6]^{-3}$ (Δ_0 order)

(2) $[Fe(CN)_6]^{-4} < [Fe(CN)_6]^{-3}$ (Fe-C Bond length)

(3) $CN^- < CO$ (π-accepting tendency)

(4) $C_2H_4 > K[PtCl_3(C_2H_4)]$ (C-C bond length)

13) Which of the following is incorrect :-

(1) Radon obtained as a decay product of ^{226}Ra

(2) Ease of liquification He > Ne > Ar > Kr > Xe

(3) All these noble gases except Rn and Og occur in the atmosphere

(4) He is a non-inflammable and light gas

14) Which of the following elements have least difference in E^0 value of M^{+2}/M :-

(1) V & Mn

(2) Mn & Fe

(3) Fe & Co

(4) Co & Ni

15) Match List-I with List-II

List-I		**List-II**	
(a)	Fe	(i)	Highest 2nd I.E. in 3d series elements
(b)	Ni	(ii)	M in $[M(CO)_4]$ is sp^3 hybridisation
(c)	Cu	(iii)	+3 oxidation state is more paramagnetic than +2
(d)	Mn	(iv)	Shows +7 oxidation state (highest)

	(a)	**(b)**	**(c)**	**(d)**
(1)	iii	i	ii	iv
(2)	iii	ii	i	iv
(3)	i	iii	ii	iv
(4)	iii	i	iv	ii

(1) 1

(2) 2

(3) 3

(4) 4

16) Unit of k for the rate law, $r = k[A][B]^{0.5}$ are :-

(1) $mol^{1/2}\ L^{-1/2}\ S^{-1}$

(2) $mol^{1/2}\ L^{1/2}\ S^{-1}$

(3) $mol^{-1/2}\ L^{+1/2}\ S^{-1}$

(4) $mol^{1/2}\ L^{1/2}\ S^{+1}$

17) For a first order reaction, $X \rightarrow Y + Z$, in the beginning of reaction 10gm of X were taken, then

what is the amount of X used in reaction after 6 half-lives ?

(1) 1.562 gm
(2) 9.844 gm
(3) 0.312 gm
(4) 0.156 gm

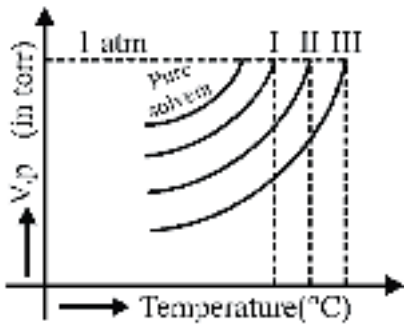
18) In the reaction, $2H_2O_2 \rightarrow 2H_2O + O_2$ The rate of decomposition of H_2O_2 is 0.68 kg hr^{-1}. What is the rate of formation of O_2 and H_2O respectively in kg hr^{-1} :-

(1) 0.36, 0.32 kg hr^{-1}
(2) 0.32, 0.36 kg hr^{-1}
(3) 0.34, 0.68 kg hr^{-1}
(4) 0.68, 0.34 kg hr^{-1}

19) If two liquids A and B have $P_A^\circ : P_B^\circ = 1 : 2$ and have mole fraction in solution in the ratio 1 : 2 then mole fraction of A in vapour phase :-

(1) 0.33
(2) 0.25
(3) 0.20
(4) 0.52

20) The vapour pressure curves of the same non-volatile, non-electrolyte solute in the same solvent are shown. The concentrations of solutions are in order of :



(1) I < II < III
(2) I = II = III
(3) I > II > III
(4) I > III > II

21) Match the column-I and column-II.

	Column-I (Cell)		**Column-II (Electrolyte used)**
(a)	Leclanche cell	(p)	KOH + ZnO
(b)	Mercury cell	(q)	$NH_4Cl + ZnCl_2$

(c)	Lead Storage battery	(r)	Moist NaOH or KOH
(d)	Ni-Cd Cell	(s)	38% solution of sulphuric acid

(1) a-p, b-r, c-q, d-s

(2) a-s, b-p, c-q, d-r

(3) a-r, b-q, c-s, d-p

(4) a-q, b-p, c-s, d-r

22) Match the column-I with column-II

	Column-I (Solution)		**Column-II (Product of electolysis)**
(P)	aq. NaCl (Pt electrode)	(i)	Anode ⇒ Cl_2 Cathode ⇒ Sodium amalgam
(Q)	dilute H_2SO_4 (Pt electrode)	(ii)	Anode ⇒ Cl_2 Cathode ⇒ Na
(R)	aq. NaCl (Hg electrode)	(iii)	Anode ⇒ O_2 Cathode ⇒ H_2
(S)	Molten NaCl (Pt electrode)	(iv)	Anode ⇒ Cl_2 Cathode ⇒ H_2

(1) P→(i); Q→(ii); R→(iv); S→(iii)

(2) P→(i); Q→(ii); R→(iii); S→(iv)

(3) P→(iv); Q→(iii); R→(i); S→(ii)

(4) P→(iv); Q→(iii); R→(ii); S→(i)

23) Match the following process with quantity of charge required

Column-I		**Column-II**	
(A)	Electrolysis of H_2O to give 5.6L O_2 at STP	(P)	3F
(B)	Conversion of 0.5 mole $Cr_2O_7^{2-}$ into Cr^{3+}	(Q)	4F
(C)	Reduction of 18g Al^{3+} into Al	(R)	1F
(D)	Electrolysis of 2L, 1M $CuSO_4$ solution to give Cu	(S)	2F

(1) A → Q, B → P, C → S, D → R

(2) A → S, B → R, C → Q, D → P

(3) A → R, B → P, C → S, D → Q

(4) A → P, B → S, C → R, D → Q

24) Which of the following statements are correct :
(i) Freshly prepared precipitate sometimes gets converted into colloidal solution due to peptization.
(ii) Both As_2S_3 sol and $Fe(OH)_3$ sol are negatively charged.
(iii) Lyophilic sol are highly solvated as particles have great affinity for solvent.
(iv) Lyophobic sols are reversible sols so special methods are required to prepare them.

(1) Only i
(2) i and iii
(3) ii and iv
(4) All of these

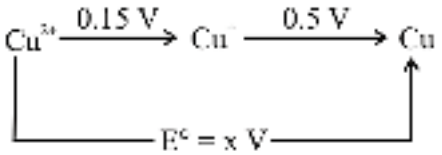
25) Molar mass of Mohr's salt is 392. In Mohr's salt iron is oxidised by $KMnO_4$ in acidic medium. Equivalent weight of Mohr's salt is :

(1) 392
(2) 31.6
(3) 278
(4) 156

26) Metals X, Y, Z have standard oxidation potentials –0.65V, –1.40V and +0.36V respectively then:-

(1) Z can oxidise X but not Y.
(2) Y can oxidise X but not Z.
(3) Z can reduce $X^{\oplus}$ and $Y^{\oplus}$ both.
(4) $X^{\oplus}$ can reduce Z but not Y.

27) In the diagram given below the value of x is :-



(1) 0.325 V
(2) 0.65 V
(3) –0.35 V
(4) –0.65 V

28) The emf of cell
$Ni_{(s)}|Ni^{+2}$ (0.16M)|| Ag^{+} (0.002M)|$Ag_{(s)}$ is
(Given that $E°_{cell}$ = 1.05 V) :-

(1) –0.91 V
(2) +0.46 V
(3) +0.91 V
(4) –0.75 V

29) What would be the freezing point of 20% ionized 0.2 molal solution of weak electrolyte XY :-
[Given : $K_f(H_2O) = 1.86°$ C/m]

(1) –0.31° C

(2) –0.45° C

(3) –0.53°C

(4) –0.90° C

30) 17.4 % (mass / volume) K_2SO_4 solution at 27°C is isotonic to 5.85% (mass / volume) NaCl solution at 27°C. If NaCl is 100% ionised, what is % ionisation of K_2SO_4 in aq. solution ?
[At wt. of K = 39, Na = 23] :-

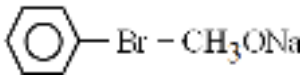
(1) 25%

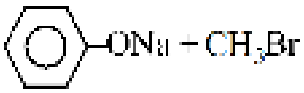
(2) 75%

(3) 50%

(4) 100%

31) Which of the reactants given below is/are suitable for the preparation of Methoxybenzene ?

(x) 

(y) 

(1) Only x

(2) Only y

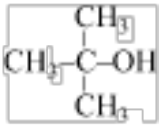
(3) Both x and y

(4) Neither 'x' nor 'y'

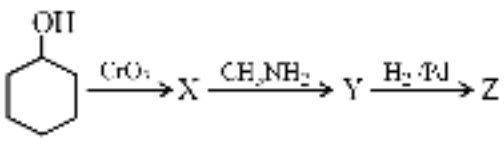
32) Which does not give dichromate test ?

(1) CH_3CH_2OH

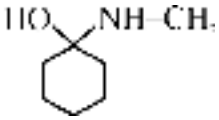
(2) 

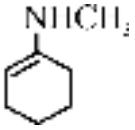
(3) 

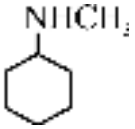
(4) CH_3OH

33) 
Product [Z] of given reaction is :-

(1) 

(2) 

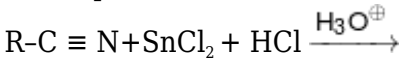
(3) 

(4) 

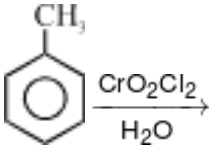
34)

In which of the following reaction aldehyde will form?

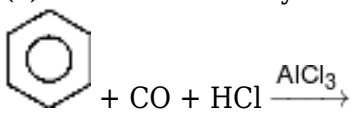
(a) Stephen's reduction

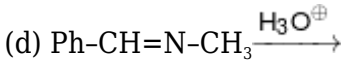


(b) Etard reaction



(c) Gattermann formylation



(d) 

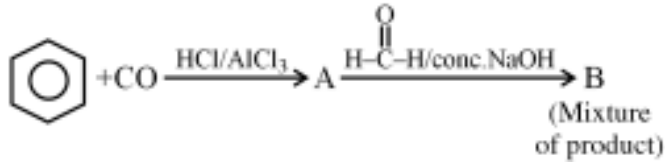
(1) Only a & b

(2) Only a & c

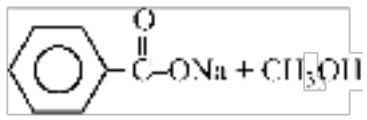
(3) Only a, c & d

(4) All of these

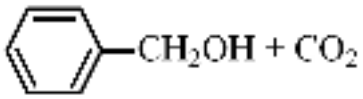
35) Suppose the following reaction.

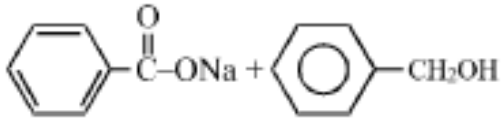


The dominant component of product will be :-

(1) 

(2) 

(3) 

(4) 

36) Identify x and y in the following sequence

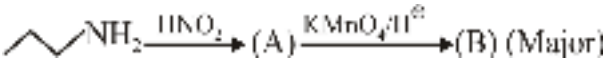
$C_2H_5Br \xrightarrow{x}$ product $\xrightarrow{y} C_3H_7NH_2$

(1) x = KCN , y = $LiAlH_4$

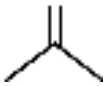
(2) x = KCN, y = H_3O^+

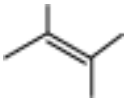
(3) x = CH_3Cl, y = $AlCl_3$/HCl

(4) x = CH_3–NH_2, y = HNO_2

37) 

B is ozonolysis product of :-

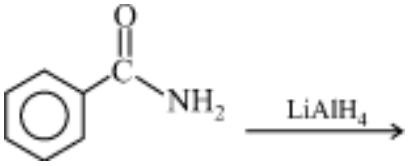
(1) 

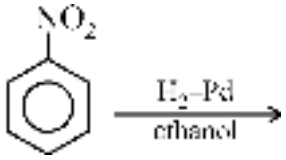
(2) 

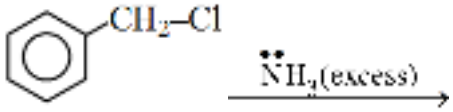
(3) 

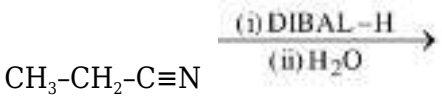
(4) (1) and (2) both

38) Which of the following reaction will not produce 1° amine ?

(1) 

(2) 

(3) 

(4) 

39) Which of the following amine can be prepared by Gabriel Phthalimide synthesis ?

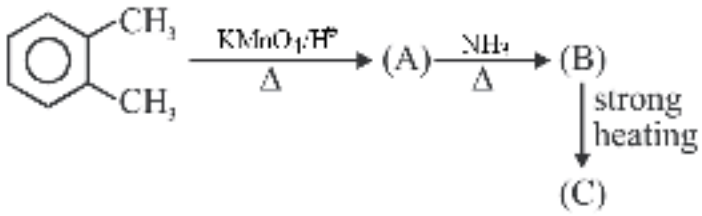
(1) Dimethyl Amine

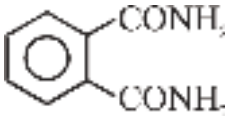
(2) Aniline

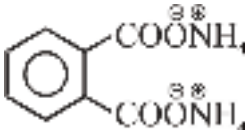
(3) Ethyl amine

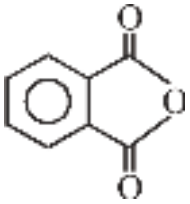
(4) Trimethyl amine

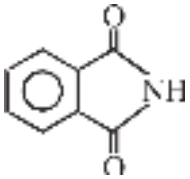
40) The product (C) will be :-

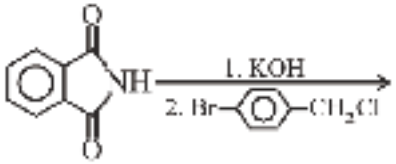


(1) 

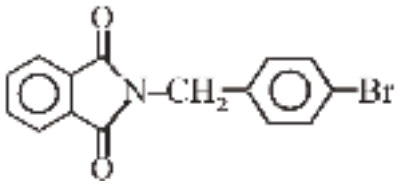
(2) 

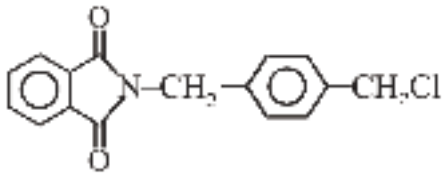
(3) 

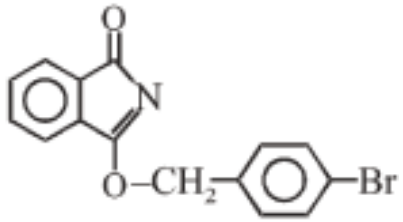
(4) 

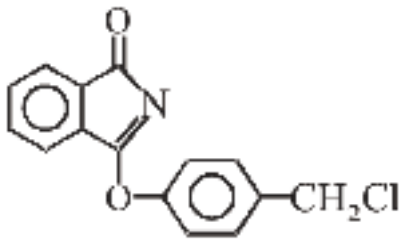
41) 

Product :-

(1) 

(2) 

(3) 

(4) 

42) In a Lassaigne's test for sulphur in the organic compound with sodium nitroprusside solution the purple colour formed is due to-

(1) $[Fe(CN)_5\ NOS]^{4-}$

(2) $[Fe(CN)_5\ S]^{2-}$

(3) $[Fe(CN)_5\ NOS]^{2-}$

(4) $[Fe(CN)_6\ ]^{4-}$

43) Which of the following alcohol does not give a red color in victor meyer test :-

(1) Isobutyl alcohol

(2) Diethyl carbinol

(3) Phenyl carbinol

(4) Ethanol

44) Which of the following does not react with phenol ?

(1) Na metal

(2) Sodium hydroxide

(3) Potassium hydroxide

(4) Sodium bicarbonate

45) Consider thiol anion ($RS^{\ominus}$) and alkoxy anion ($RO^{\ominus}$). Which of the following statement is correct ?

(1) $RS^{\ominus}$ is less basic but more nucleophillic than $RO^{\ominus}$

(2) $RS^{\ominus}$ is more basic and more nucleophillic than $RO^{\ominus}$

(3) $RS^{\ominus}$ is more basic but less nucleophilic than $RO^{\ominus}$

(4) $RS^{\ominus}$ is less basic and less nucleophillic than $RO^{\ominus}$

BIOLOGY

1) Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): A flower is defined as modified shoot wherein the shoot apical meristem changes to floral meristem.

Reason (R) : Internode of the root gets condensed to produce different floral appendages laterally at successive nodes instead of leaves.
In the light of the above statements, choose the correct answer from the options given below :

(1) Both A and R are true, but R is NOT the correct explanation of A.
(2) A is true, but R is false.
(3) A is false, but R is true.
(4) Both A and R are true, and R is the correct explanation of A.

2) Which statement is not correct :

(1) In grasses, the guard cells are dumb-bell shaped
(2) Subsidiary cells are not a part of stomatal apparatus
(3) On the stem the epidermal hairs are called trichomes
(4) The root hairs are unicellular elongations of the epidermal cell

3) Find the incorrect match from the following :

(1) Monocot stem - Radial vascular bundle
(2) Dicot root - Exarch primary xylem
(3) Dicot stem - Endarch primary xylem
(4) Dicot leaf - Parenchymatous bundle sheath

4) Mango and coconut are example of which type of fruits ?

(1) Berry
(2) Pome
(3) Nut
(4) Drupe

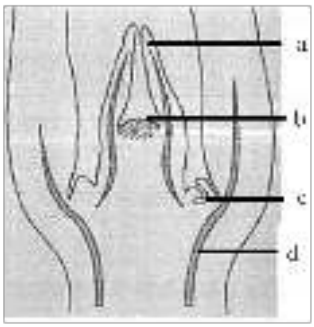
5) Which type of symmetry is found in flowers of mustard ?

(1) Zygomorphic
(2) Actinomorphic
(3) Asymmetrical
(4) 1 and 3

6) Sclerenchymatous hypodermis is found in :

(1) Monocot root
(2) Monocot stem
(3) Dicot root
(4) Dicot stem

7) Recognise the figure and find out the correct matching :



(1) a-axillary bud, b-differentiating vascular tissue, c-leaf primordium, d-shoot apical meristematic zone

(2) a-differentiating vascular tissue, b-axillary bud, c-shoot apical meristematic zone, d-leaf primordium

(3) a-leaf primordium, b-shoot apical meristematic zone, c-axillary bud, d-differentiating vascular tissue

(4) a-shoot apical meristematic zone, b-leaf primordium, c-differentiating vascular tissue, d-axillary bud

8) In monocot stem, peripheral vascular bundle are generally :

(1) Larger than the centrally located ones.

(2) Smaller than the centrally located ones

(3) Similar than centrally located ones

(4) Larger but non functional

9) **Statement I** - Coconut water of tender coconut is the free nuclear endosperm.
Statement II- The white kernel surrounding it represents the cellular endosperm.

(1) Both statement I and II are correct

(2) Statement I correct, Statement II incorrect

(3) Statement I incorrect, Statement II correct

(4) Both Statement I and II are incorrect

10) Which of the following correctly describes the structure of a typical dicot embryo?

(1) It has one cotyledon, epicotyl ends in radicle and hypocotyl ends in plumule

(2) It has two cotyledons, epicotyl ends in plumule and hypocotyl ends in radicle covered by root cap.

(3) It has two cotyledons, hypocotyl ends in plumule and radicle forms epicotyl.

(4) It has many cotyledons in which radicle and plumule are absent

11)

Column I (Structure)	**Column II (Description)**
(A) Scutellum	1. Upper part of embryonal axis containing the shoot apex and leaf primordia

(B) Coleorhiza	2. Lateral cotyledon found in members of the grass family
(C) Epicotyl	3. Undifferentiated sheeth enclosing the radicle and rootcap.
(D) Coleoptile	4. Hollow foliar structure covering the epicotyl
(E) Radicle+Root cap	5. Lower end of the emryo which form primary root

(1) A-1 B-3 C-2 D-5 E-4

(2) A-2 B-3 C-1 D-4 E-5

(3) A-2 B-1 C-3 D-5 E-4

(4) A-2 B-4 C-3 D-1 E-5

12) **Assertion (A) :** In a majority of flowering plants monosporic development of embryo sac occurs.
Reason (R) : Out of four megaspores form from one MMC, only one is functional while the other three degenerate and functional megaspore develops into the female gametophyte.

(1) Assertion (A) and Reason (R) are true, and R is the correct explanation of A.

(2) Assertion (A) and Reason (R) are true, but R is not the correct explanation of A.

(3) Assertion (A) is true, but Reason (R) is false

(4) Assertion (A) is false, but Reason (R) is true.

13) Which of the following is incorrect ?

(1) Six of the eight nuclei of embryo sac are surrounded by cell walls and organised into cells

(2) Two nuclei, called polar nuclei are situated below the egg apparatus in the large centralcell

(3) Three cells are grouped together at the micropylar end and constitute the egg apparatus

(4) The egg cells have special cellular thickenings at the micropylar tip called filiform apparatus

14) **Statement I :** Cleistogamous flowers produce assured seed-set even in the absence of pollinators.
Statement II : Cleistogamous flowers are invariably autogamous because of close flowers.

(1) Both statements are correct

(2) Statement I is correct & II is incorrect

(3) Statement I is incorrect & II is correct

(4) Both statements are incorrect

15) Find the correct option with respect to Xenogamy :-
(a) Transfer of pollen grains from anther to the stigma of a different plant
(b) During this pollintion genetically similar types of pollen grains reach to the stigma.
(c) This is the only type of pollination which during pollintion brings genetically different types of pollen grains to the stigma.
(d) Transfer of pollen grains from anther to the stigma of a same plant

(1) (c) and (d) Only

(2) (a) and (d) Only

(3) (b) and (d) Only

(4) (a) and (c) Only

16) Which of the following statements is incorrect with respect to sickle-cell anaemia?

(1) Substitution of amino acid in the globin protein results due to single base substitution at the sixth codon of the beta globin gene from GAG to GUG.

(2) The defect is caused by the substitution of valine (val) by glutamic acid (Glu).

(3) The mutant haemoglobin molecule undergoes polymerization under low oxygen tension.

(4) Shape of RBC change from biconcave disc to elongated sickle like structure.

17) Calculate the probability of a pea plant of being dwarf in F_1 when homozygous tall and dwarf plants are crossed together?

(1) 1/4

(2) 1/2

(3) 0

(4) 1

18) Match the Column-I and Column-II w.r.t chromosomal disorder and select the correct option.

	Column-I		**Column-II**
A	Down's syndrome	(i)	Gynaecomastia
B	Turner's syndrome	(ii)	Short statured with small round head, furrowed tongue and partially open mouth
C	Klinefelter's syndrome	(iii)	Sterile female with underdeveloped feminine characters

	A	**B**	**C**
(1)	(ii)	(iii)	(i)
(2)	(i)	(ii)	(iii)
(3)	(iii)	(ii)	(i)
(4)	(i)	(iii)	(ii)

(1) 1

(2) 2

(3) 3

(4) 4

19) Select the incorrect statement w.r.t 'Haemophilia'.

(1) It is a recessive disorder.

(2) Shows its transmission from carrier male to some of the female progeny.

(3) In this disease, a single protein that is a part of the cascade of proteins involved in clotting of blood is affected.

(4) The possibility of female becoming a haemophilic is extremely rare.

20) Morgan selected *Drosophila as* experimental material because of all of the given reasons, except :-

(1) *Drosophila* could be grown on simple synthetic medium in the laboratory

(2) *Drosophila* completes its life cycle in about two weeks.

(3) *Drosophila* grows on fruits.

(4) *Drosophila* has many types of hereditary variations that can be seen with low power microscope.

21) Match the columns I and II, and choose the correct combination from the options given.

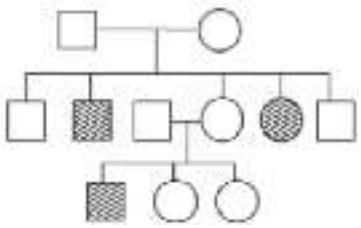
	Column-I		**Column-II**
a.	XO type	1.	Male heterogamety
b.	XY type	2.	Female heterogamety
c.	ZW type		

(1) a - 1, b - 2, c - 2

(2) a - 2, b - 1, c - 2

(3) a - 1, b - 1, c - 2

(4) a - 2, b - 2, c - 1

22) Study the pedigree chart given below :-  What does it show :-

(1) Inheritance of a recessive sex-linked disease like haemophilia

(2) Inheritance of a sex-linked inborn error of metabolism like phenylketonuria

(3) Inheritance of a condition like phenylketonuria as an autosomal recessive trait

(4) The pedigree chart is wrong as this is not possible

23) In a double stranded DNA sample, 20 molecules of pentose sugar are present. In this sample, what will be the number of purine nitrogen bases?

(1) 10

(2) 5

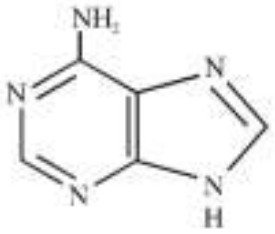
(3) 6

(4) 20

24) **Assertion:** The pitch of the DNA helix is 3.4 nm.

Reason: The distance between 2 base pairs is 0.34 nm & there are 10 bp in each turn.

(1) If both Assertion & Reason are True & the Reason is a correct explanation.
(2) If both assertion & reason are true but reason in not a correct explanation of the assertion
(3) If Assertion is true but the reason is false
(4) If both Assertion & Reason are false



25) The following structure is related to which compound?

(1) Cytosine
(2) Adenine
(3) Uracil
(4) Thymine

26) Which is correct with regard to mature mRNA in eukaryotes ?

(1) Exons and introns do not appear in the mature RNA.
(2) Exons appear but introns do not appear in the mature RNA.
(3) Introns appear but exons do not appear in the mature RNA.
(4) Both exons and introns appear in the mature RNA.

27) In the lac operon model, lactose molecules function as:

(1) Inducer which bind with operator gene
(2) Repressor which bind with operator gene
(3) Inducer which bind with repressor protein
(4) Corepressor which bind with repressor protein

28) Find out correct statement about transcription:-

(1) In transcription of DNA only one strands is copied into RNA.
(2) In all eukaryotes single DNA polymerase is capable of catalyse all the three steps of transcription, which are initiation, elongation and termination.
(3) In all prokaryotes transcription and translation take place in the different compartment.
(4) In all eukaryotes primary transcript contain only exons.

29) In the process of transcription in eukaryotes, the RNA polymerase II transcribes

(1) Precursor of mRNA
(2) tRNA and SnRNAs
(3) 5sRNA

(4) 28S, 5.8Sand 18S rRNA

30) **Assertion :** The relationship between genes and DNA are best understood by mutation studies
Reason : Frame shift mutation forms the genetic basis of proof that codon is a triplet and it is read in a contiguous fashion.

(1) Both assertion and reason are true and the reason is a correct explanation of the assertion.

(2) Both assertion and reason are true but reason is not a correct explanation of the assertion.

(3) Both assertion and reason are false.

(4) Both assertion and reason are true.

31) **Statement-I :** In an aquatic ecosystem GFC is major conduit of energy flow.
Statement-II : In a terrestrial ecosystem DFC is major conduit of energy flow.

(1) Both Statement-I and Statement-II are incorrect

(2) Statement-I is correct but Statement-II is incorrect

(3) Statement-I incorrect and Statement-II is correct

(4) Both Statement-I and Statement-II are correct

32) Read the following five statements about ecological pyramid :
(a) Food chain is used to build ecological pyramid.
(b) Saprophytes are places on top at the pyramid.
(c) Pyramids of grassland ecosystem are always upright.
(d) Pyramids of biomass in aquatic ecosystem are mostly inverted.
(e) Concept of ecological pyramids was given by Charles Darwin.
Choose the option which includes correct statements only.

(1) b and e

(2) a, c and d

(3) c,d and e

(4) a, b, c and e

33) Which one of the following processes during decomposition is correctly described?

(1) Leaching –Water soluble inorganic nutrients rise to the top layers of soil

(2) Fragmentation – Carried out by organisms such as earthworm

(3) Humification – Leads to the accumulation of a dark coloured substance humus which undergoes microbial action at a very fast rate

(4) Catabolism – Last step in the decomposition under fully anaerobic condition

34) The rate of decomposition is faster in the ecosystem due to following factors EXCEPT:

(1) Detritus rich in sugars

(2) Warm and moist environment

(3) Presence of aerobic soil microbes

(4) Detritus richer in lignin and chitin

35) Which one of the following reasons is proposed for the greater species diversity in tropical environments compared to temperate ones ?

(1) Less solar energy in tropics

(2) More frequent glaciations in tropics

(3) Relatively constant environments in tropics

(4) Shorter evolutionary time in tropics

36) When we remove the barnacle *Balanus* from the rocky intertidal regions of Scotland then the barnacle *Chathamalus* increase in number and cover more area. This is the example of which of the following

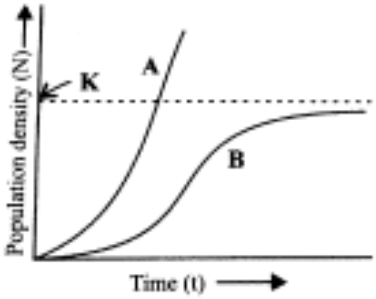
(1) Amensalism

(2) Competition

(3) Resource partitioning

(4) Competitive release

37) Given below is the representation of population growth models/curve.



Choose correct option w.r.t. A and B

(1) A ⇒ Exponential / Geometric / J shape curve when resources are limited.

(2) $B \Rightarrow \frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$

(3) A ⇒ Logistic / Sigmoid / S–shape curve when resources are limited

(4) $B \Rightarrow \frac{dN}{dt} = rN$ or $\frac{dN}{dt} = (b - d)N$ or $N_t = N_o e^{rt}$

38) **Statement I :** Predators acts as conduit for energy transfer across trophic levels.
Statement II : The prickly pear cactus introduced into Australia in the early 1920's caused havoc by spreading rapidly into millions of hectares of rangeland.
In the light of above statements.
Choose the most appropriate answer from the option given below :

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

39) In the following statement choose wrong one -
(a) Wasp laying egg in a fig fruit is an example of mutual relationship.
(b) Mycorrhiza is an example of ectoparasitism.

(c) Physiological ecology is ecology at organism level.
(d) Competitive exclusion principle was given by connell.

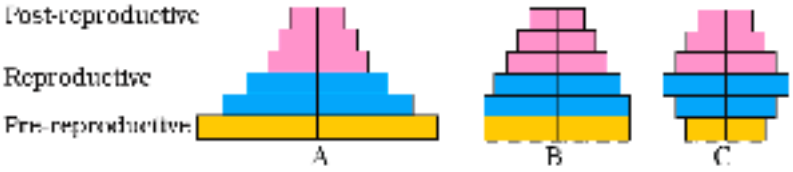
(1) a & c

(2) a, c, d

(3) b and d

(4) b, c and d

40) What does the shape of the given age pyramids (A to C) reflect about the growth status of populations?



	A	B	C
(1)	Declining	Stable	Expanding
(2)	Stable	Expanding	Declining
(3)	Expanding	Stable	Declining
(4)	Declining	Expanding	Stable

(1) 1

(2) 2

(3) 3

(4) 4

41) **Assertion (A):** Habitat loss is the most important cause of biodiversity extinction.
Reason (R): Urbanization and deforestation directly destroy natural ecosystems.

(1) A and R are true, and R explains A.

(2) A and R are true, but R does not explain A.

(3) A is true, R is false.

(4) A is false, R is true.

42) Which of the following statements about overexploitation of resources are true?
A. It leads to decline in population of species
B. It is responsible for extinction of the passenger pigeon
C. Medicinal plants are unaffected by overexploitation.
D. Deforestation is a form of overexploitation.
Options:

(1) A, B and D only

(2) A and C only

(3) B and C only

(4) All of the above

43)

Match the column

Column-I (Cause)		Column-II (Example)	
(a)	Habitat loss	(i)	Amazon rainforest deforestation
(b)	Alien species invasion	(ii)	Introduction of Nile perch in Lake Victoria
(c)	Co-extinction	(iii)	Extinction of host leads to parasite loss
(d)	Over exploitation	(iv)	Overfishing of marine fish

Options:

(1) a-(i), b-(ii), c-(iii), d-(iv)

(2) a-(ii), b-(i), c-(iv), d-(iii)

(3) a-(iii), b-(iv), c-(ii), d-(i)

(4) a-(iv), b-(iii), c-(i), d-(ii)

44) Find out the correct one-

(1) India has 7.1% species diversity

(2) India is one of the 22 mega biodiversity countries of world

(3) According to May's global estimate only 22% of the total species have been recorded so for

(4) According to may's estimation, more than 3,00,000 plant species yet to be discovered.

45) Which of the following statements are true about ecological succession ?
(a) The members of pioneer community are those who have more nutrient demand.
(b) Usually mesophytes are members of climax community.
(c) Secondary succession begins in areas where there was previous vegetation present.
(d) The climax community is not predictable.

(1) a, b

(2) b, c

(3) a, d

(4) a, c

46) Which of the following statements are correct ?
(A) In PCR, RNA primers are synthesised by primase.
(B) Cry I Ab control the cotton bollworms
(C) In PCR denaturation of double stranded DNA is induced by high temperature.
(D) There are an estimated 200,000 documented varieties of basmati rice in India.
Choose the most appropriate answer from the options given below.

(1) A, B and C only

(2) A, B and D only

(3) Only B and C

(4) Only C

47) Read the following statements carefully and select the incorrect statements.
(I) Bt toxin becomes activated in acidic conditions inside the stomach of mammals and exerts toxic effects on them.
(II) Cry proteins destroy insect larvae by directly degrading nuclear DNA of midgut epithelial cells.
(III) RNA interference (RNAi) involves single-stranded RNA directly binding to DNA and permanently blocking transcription.
(IV) Genetic modifications introduced in somatic gene therapy are inherited to the next generation.
(V) Transgenic plants are developed exclusively to enhance crop yield and are not used for pest resistance or nutritional enhancement.
(VI) Bt gene is derived from a plant virus that infects cotton plants.
Options :

(1) (I), (II), (III), (IV), (V) and (VI) are incorrect

(2) (I), (III), (IV) and (VI) are incorrect

(3) (II), (IV) and (V) are incorrect

(4) (I) and (VI) are incorrect

48) Identify the blank spaces A, B, C and D in the following given table and select the **correct** options :

Type of microbe	Scientific name	Commercial product
Bacterium	A	Streptokinase
B	Aspergillus niger	Citric acid
Fungus	Trichoderma polysporum	C
Bacterium	D	Butyric acid

(1) A-*Streptococcus*. B-Fungus, C-Cyclosporin-A, D-*Clostridium butylicum*

(2) A-*Clostridium butylicum*, B-*Streptococcus*. C-Fungus. D-Cyclosporin-A

(3) A-*Streptococcus*, B-Yeast, C-Cyclosporin-A, D-*Lactobacillus*

(4) A-*Streptococcus*, B-Cyclosporin-A, C-Sratins, D-*Clostridium butylicum*

49) **Assertion:** Penicillin antibiotic was discovered by Alexander fleming from a mould.
Reason: Alexander fleming while working on *Staphylococci* bacteria, once observed that a mould growing in one of his unwashed culture plates around which *Staphylococci* could not grow.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

50) Which of the following statement is correct ?

(1) Genetic engineering approval committee was establised by UNO (United Nations Organisation)

(2) There are an estimated 2000 varieties of rice in India alone.

(3) There are 47 documented varieties of basmati rice grown in India.

(4) Basmati rice is distinct for its unique aroma and flavour.

51) **Assertion :-** The most commonly used bioreactors are of stirring type.
Reason :- The stirrer facilitates even mixing and oxygen availability throughout the bioreactor.

(1) Both assertion and reason are true but reason is not correct explanation of assertion

(2) Both assertion and reason are true and reason is correct explanation of assertion

(3) Assertion is true but reason is false

(4) Assertion is false but reason is false

52) Select the incorrect statement :-

(1) Hind-II was the Ist restriction endonuclease enzyme discovered.

(2) In EcoRI enzyme, 'R' represent the Genus of organism.

(3) Each restriction endonuclease recognizes specific palindromic nucleotide sequence.

(4) Palindromic sequence of EcoRI is 5'GAATTC3', 3'CTTAAG5'

53)

Ti plasmid transforms cell of

(1) Animals

(2) Plants

(3) Bacteria

(4) Fungi

54)



Which of the following pairs of restriction enzymes have restriction site within amp^R and tet^R genes respectively in pBR322 plasmid ?

(1) Pvu II, BamH I

(2) Pst I, Sal I

(3) Hind III, EcoR I

(4) Cla I, Pst I

55) The DNA molecules to which the gene of interest is integrated for cloning is called :

(1) Carrier

(2) Transformer

(3) Vector
(4) Template

56) Which one of the following technique is used to produce the GM crops ?

(1) Micropropagation
(2) Somatic hybridization
(3) r-DNA technology
(4) Cross breeding

57) Bacterium *Bacillus thuringiensis* is widely used in contemporary biology as -

(1) Source of industrial enzyme
(2) Indicator of water pollution
(3) Microbial biocontrol agent
(4) Agent for production of dairy products.

58) The Indian parliament has recently cleared the ____ amendment of the Indian patents Bill that takes such issues into consideration, including patent terms emergency provisions and research and development initiative :-

(1) First
(2) Third
(3) Second
(4) Fifth

59) Which one of the following is an example of carrying out biological control of pest using fungi?

(1) *Bacillus thuringiensis* against butterfly caterpillars
(2) Baculoviruses against certain insects and arthropods.
(3) Lady bird beetle against aphids
(4) *Trichoderma* species against certain plant pathogens.

60) In the given table some structures related to the human reproductive system and their functions are given. Fill in the blanks in the table by using the given options.

Structures	**Functions**
Seminiferous tubules	Sperms production site
__(i)__	Nutrition to germ cells
Leydig cell	__(ii)__
__(iii)__	Lubrication of the penis

(1) (i) - Sertoli cell, (ii) - Secretes androgen, (iii) - Seminal vesicle

(2) (i) - Sertoli cell, (ii) - Secretes androgen,
(iii) - Bulbourethral gland

(3) (i) - Spermatogonia, (ii) - Secretes androgen,
(iii) - Bulbourethral gland

(4) (i) - Golgi body, (ii) - Secretes estrogen,
(iii) - Seminal vesicle

61) Read the given passage and answer the following question.
In the menstrual cycle of a human female, a temporary endocrine structure 'P' is formed in the ovary, which secretes a large amount of hormone 'Q'. Hormone 'Q' is essential for the maintenance of layer 'R' of uterus. Implantation of blastocyst occurs in layer 'R'.
Arrange all the events in the correct sequence during the formation of 'P' and select the correct option.
(W) - Extreme high concentration of estrogen
(X) - Formation of corpus luteum
(Y) - FSH and LH surge
(Z) - Rupturing of graafian follicle

(1) Z → Y → W → X

(2) Z → Y → X → W

(3) Y → X → W → Z

(4) W → Y → Z → X

62) In a normal pregnant woman, the amount of gonadotropin was assessed. The result expected was :-

(1) High levels of FSH and LH in uterus to stimulate endometrial thickening

(2) High levels of circulating HCG to stimulate estrogen and progesterone synthesis

(3) High level of circulating FSH and LH in the uterus to stimulate implantation of the embryo

(4) High level of circulating HCG to stimulate endometrial thickening

63) **Assertion (A) :** Ovaries are primary female sex organs that produce the female gamete and several steroid hormones.
Reason (R) : Placenta acts as an endocrine tissue and produces several hormones.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

64) Read the following two statements and choose the correct option :
Statement-I : The uterus opens into cervix through a narrow vagina.
Statement-II : The cavity of the cervix is called cervical canal which along with vagina forms the birth canal.

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect.

(4) Statement I is incorrect but Statement II is correct.

65) Choose the incorrect statement regarding female reproductive system.

(1) Oogenesis is initiated during the embryonic development stage.
(2) Primary oocytes are arrested in prophase-I of *meiosis* until puberty.
(3) All oogonia complete meiosis before birth.
(4) The second polar body is formed only if fertilization occurs.

66) Funnel shaped structure of fallopian tube is-

(1) Infundibulum
(2) Fimbriae
(3) Ampulla
(4) Isthmus

67) During spermatogenesis first haploid structure is called as :-

(1) Spermatid
(2) Spermatozoa
(3) Primary spermatocyte
(4) Secondary spermatocyte

68)

The second polar body is formed :

(1) Before entry of sperm into ovum
(2) During meiosis-I division
(3) Before occurring second meiotic division
(4) After entry of sperm into ovum but before completion of fertilisation

69) Cleavage divisions starts as zygote moves through _____ towards uterus.

(1) Ampulla
(2) Isthmus
(3) Infundibulum
(4) Cervix

70) Read the following statements and choose the correct option
Statement A: The Medical Termination of Pregnancy (Amendment) Act, 2017 was enacted by the government of India with the intension of reducing the incidence of illegal abortion and consequent maternal mortality and morbidity. **Statement B:** Early symptoms of STDs are minor and include itching, fluid discharge, slight pain, swellings, etc., in the genital region. Infected females may often be asymptomatic and hence, may remain undetected for long.

(1) Both the statements are correct

(2) Both the statements are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

71) Match contraceptive methods with their mode of action.

	Column I		**Column II**
a.	Vasectomy	(i)	prevents fertilisation
b.	Copper-T	(ii)	Suppresses sperm motility
c.	Condom	(iii)	Inhibits ovulation
d.	Oral pills	(iv)	Prevents sperm transport

(1) a-iv, b-ii, c-i, d-iii

(2) a-i, b-ii, c-iii, d-iv

(3) a-iv, b-i, c-ii, d-iii

(4) a-iv, b-i, c-iii, d-iv

72) Given below are two statements: **Statement I :** The term 'reproductive health' simply refers to healthy reproductive organs with normal functions.
Statement II : Amniocentesis is used to test for the presence of certain STD's
Choose the correct answer from the options given below.

(1) Both **Statement I** and **Statement II** are correct

(2) Both **Statement I** & **Statement II** are incorrect

(3) **Statement I** is correct but **Statement II** is incorrect

(4) **Statement I** is incorrect but **Statement II** is correct

73) Both the male and the female _____ are disposable, can be self-inserted and there by gives privacy to the user.

(1) Diaphragms

(2) Cervical caps

(3) Vaults

(4) Condoms

74) Embryo with upto 8 blastomeres formed due to in vitro-fertilisation is transferred into

(1) Uterus

(2) Fallopian tube

(3) Cervix

(4) Vagina

75) In a random mating population of 5000 individuals frequency of dominant phenotype (character shows complete dominance) is 0.51, then find the
(i) Number of homozygous recessive individuals.

(ii) Number of homozygous individuals

(1) i - 2450, ii - 450
(2) i - 2550, ii - 2900
(3) i - 2450, ii - 2900
(4) i - 2450, ii - 2900

76) **Statement-I :** Natural selection is a process in which heritable variable enabling better survival are enabled to reproduce and leave greater number of progeny.
Statement-II : Evolution by natural selection, in a true sense would have started when cellular forms of life with differences in metabolic capability originated on earth.

(1) Statement-I is correct and statement-II is incorrect.
(2) Statement-I is incorrect and statement-II is correct.
(3) Both statements are incorrect.
(4) Both statements are correct.

77) Match the following types of evidences with their correct examples.

	Evidences		**Examples**
A	Paleontological evidence	i	Similarity in vertebrate embryos during early stages
B	Anatomical evidences	ii	Presence of fossils in sedimentory rocks
C	Embryological evidence	iii	Similarity in DNA and protein sequences among different organisms
D	Molecular evidence	iv	Presence of homologous structures like forelimbs of different vertebrates
E	Biogeographical evidences	v	Unique distribution of marsupials in Australia

(1) A-ii ; B-iv ; C-i ; D-iii ; E-v
(2) A-ii; B-iii ; C-v ; D-iv ; E-i
(3) A-i ; B-iii ; C-ii ; D-v ; E-iv
(4) A-iv ; B-i ; C-iii ; D-ii ; E-v

78) According to Hugo de Vries, speciation is result of -

(1) Gradual variation
(2) Use and disuse of organ
(3) Saltation
(4) Genetic drift

79) The theory of special creation was not objected by which of the following statements ?

(1) Existing life forms share similarities to varying degrees not only among themselves but also with life forms that existed millions of years ago.

(2) Many life forms do not exist now.

(3) There has been gradual evolution of life forms.

(4) The earth was about 4000 years ago.

80) Which statement correctly describes mammals?

(1) First mammals were large sized and oviparous

(2) Mammals protect their unborn young inside the mother's body

(3) Mammals dominated earth before reptiles

(4) All mammals live only on land

81) **Assertion (A) :** HIV infection leads to a progressive decline in the functioning of the immune system because the virus specifically attacks T-helper cells causing a gradual reduction in their number and weakening the body's defense mechanisms.
Reason (R) : T helper cells co-ordinate the activities of various immune cells such as macrophages, T-killer cells and B-lymphocytes and their loss severely disrupts both cell mediated and antibody mediated immunity.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Assertion is false but Reason is true.

82) Which of the given disease is correctly matched with its correct category?
(A) Malaria, Ascariasis – Protozoan disease
(B) Hepatitis-B, AIDS – STDs
(C) Pneumonia, Typhoid – Bacterial disease
(D) Filariasis, Amoebiasis – Helminthic disease
(E) Common cold, Dengue – Viral disease

(1) A, B & D

(2) A, B & E

(3) B, C & E

(4) A, C & D

83) Find the correct option related to the name of the plant shown in the picture and its effects on



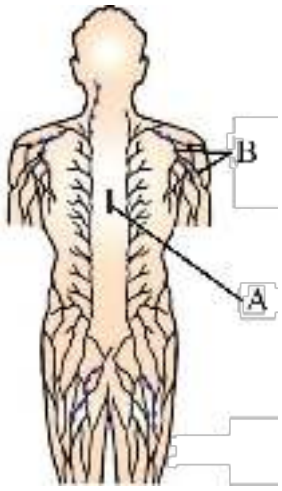
our body.

(1) Papaver somniferum - effects on cardivascular system

(2) Erythroxylum coca - potent stimulating action on central nervous system

(3) Datura - hallucinogenic properties

(4) Atropa belladona - hallucinogenic properties



84) Identify (A), (B) in following diagram and choose the correct match.

(1) (A) Thymus - Along with bone marrow provide micro environment for the development and maturation of T-lymphocytes
(B) Lymph nodes - Serve to trap the micro organism or other antigen.

(2) (A) Thymus- Serve to trap the microorganism or other antigen
(B) Lymph nodes - Provide micro environment for the development and maturation of T-lymphocyte.

(3) (A) Lymph nodes - Help in development of B lymphocyte
(B) Spleen - Maturation of lymphocytes

(4) (A) Bone marrow - Provides acquired immunity.
(B) Thymus - Serve to trap the micro organism or other antigen

85)

A newborn baby receives antibodies through the placenta before birth. After a few months, the baby is given DPT vaccine. Which of the following best describes the acquired immunity in the baby ?

(1) Baby first received natural active immunity and then artificial passive immunity.

(2) Baby first received artificial passive immunity and then natural active immunity.

(3) Baby first received natural passive immunity and then artificial active immunity.

(4) Baby received only natural active immunity in both cases.

86) Find out incorrect match from the followings:

(1) Withdrawal syndorme - characterised by anxiety, shakiness, nausea and sweating.

(2) Cannabinoids - Are a group of chemical which interact with cannabinoid receptors present principally in the brain

(3) Heroin - Heroin is a stimultant which speed-up the body functions and has hallucinogenic properties also.

(4) Malignant tumors - Are a mass of proliferatic cells or neoplastic cells, which grow rapidly, invading and damaging the surrounding normal tissues.

87) MRI uses :-

(1) X-rays
(2) Strong magnetic fields
(3) Non-ionising rays
(4) (2) & (3) both

88) Which test is carried out if disease is caused by Salmonella typhi ?

(1) Shick test
(2) Urine test
(3) Mantoux test
(4) Widal test

89) Common symptoms of allergy include:

(1) Fever and chills
(2) Sneezing and itching
(3) Vomiting only
(4) Paralysis

90) How is a cancerous cell different from a normal cell ?

(1) Cancerous cells show uncontrolled cell division, while normal cells divide in a regulated manner.
(2) Cancerous cells do not exhibit contact inhibition, while normal cells do.
(3) Both (1) and (2).
(4) Cancerous cells die immediately after abnormal division

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	2	4	3	1	4	3	1	3	4	2	3	3	2	1	3	2	1	1	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	3	1	4	3	3	3	2	1	4	2	2	3	4	3	2	2	3	2	2
Q.	41	42	43	44	45															
A.	1	3	3	2	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	1	1	1	2	1	3	1	3	1	1	4	2	1	2	3	2	2	3	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	3	3	2	1	3	1	3	2	3	2	3	4	4	2	1	4	4	3	4
Q.	86	87	88	89	90															
A.	1	1	2	4	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	2	1	4	2	2	3	2	1	2	2	1	4	1	4	2	3	1	2	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	3	1	1	2	2	3	1	1	2	4	2	2	4	3	4	2	1	3	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	1	1	3	2	4	1	1	1	4	2	2	2	2	3	3	3	3	4	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	2	2	4	3	1	4	4	2	1	1	3	4	2	3	4	1	3	4	2
Q.	171	172	173	174	175	176	177	178	179	180										
A.	1	3	3	1	3	3	4	4	2	3										

SOLUTIONS

PHYSICS

1) Relation between field and potential $E = -\frac{dV}{dr}$ or $E = -\nabla V$

$E_x = -\frac{d(xy)}{dx} = -y$

$E_y = -\frac{d(xy)}{dy} = -x$

$\vec{E} = -y\hat{i} - x\hat{j}$

(Electric field has unit N/C or V/m, not N/m)

$\tan\theta = \frac{E_y}{E_x} = \frac{x}{y}$

2)

Field at r (= x) = Field at x = 2R

$\frac{KQx}{R^3} = \frac{KQ}{(2R)^2}$

$\Rightarrow x = \frac{R}{4}$

3) $U = U_1 + U_2 = -pE\cos\theta + \frac{k(q)(-q)}{2\ell}$

$= -pE\cos\theta - \frac{kq^2}{2\ell}$

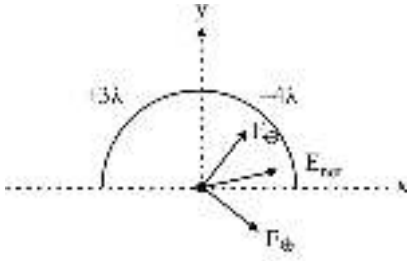
4)

For electron specific charge $\frac{e}{m}$ indicate charge per kg of electron.

Hence $\frac{e}{m} = \frac{1.6 \times 10^{-19}}{9.1 \times 10^{-31}} = 1.76 \times 10^{11}$ C/kg

Charge in 1 kg is 1.76×10^{11} C

Charge in 1 g is 1.76×10^{8} C



5)

Electric field due to +3λ at center is

$E_\oplus = \frac{\sqrt{2}k(3\lambda)}{r}$

Electric field due to –4λ at center is

$\Rightarrow E_{\ominus} = \frac{\sqrt{2}k(4\lambda)}{r}$

Resultant electric field

$E_{net} = \frac{\sqrt{2}k\lambda}{r}\sqrt{3^2+4^2}$

$E_{net} = \frac{5\sqrt{2}k\lambda}{r}$

6)

Apply KCL

$I_{ilc} = I_{oIg}$

$12 + 3 + 5 = i = 20A$

7) To convert a galvanometer into an ammeter a low value resistance is to be connected in parallel to it (called shunt).

8)

Its a balanced wheat stone bridge.

9) $R = R_0(1 + \propto \Delta t)$

$3R_0 = R_0(1 + 4 \times 10^{-3}T)$

$2 = 4 \times 10^{-3}T$

$\Rightarrow T = 1000/2 = 500°C$

10) $q = CV$

$q = \frac{\in_0 AV}{d}$

$A = \frac{qd}{\in_0 V}, V = Ed$

$A = \frac{12 \times 10^{-6}}{8.8 \times 10^{-12} \times 3 \times 10^6}$

$A = \frac{qd}{\in_0 Ed} = \frac{q}{\in_0 E}$

$A = 0.45\ m^2$

11)

Plates of parallel plate capacitor having equal and opposite charges.

12) $U_i = \left(\frac{1}{2}CV^2\right) \times 2 = CV^2$

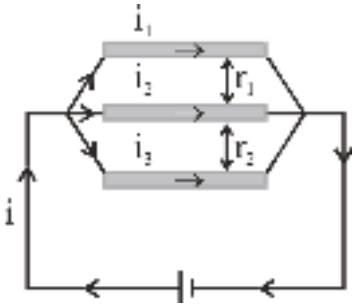
$U_f = \frac{1}{2}(3C)V^2 + \frac{(CV)^2}{2(3C)} = \frac{5}{3}CV^2$

$\frac{U_i}{U_f} = \frac{3}{5}$ Ans.

13) The wires are in parallel and ratio of their resistances are 3 : 4 : 5, Hence currents in wires are in the ratio $\frac{1}{3} : \frac{1}{4} : \frac{1}{5}$

(k : Potential difference)

$i_1 = \frac{k}{3},\ i_2 = \frac{k}{4},\ i_3 = \frac{k}{5}$



Force per unit length between top and middle wire

$F_1 = \frac{\mu_0}{4\pi}.\frac{2i_1i_2}{r_1} = \frac{\mu_0}{4\pi} \times \frac{2\left(\frac{1}{3}\right)\left(\frac{1}{4}\right)k^2}{r_1}$.

Force per unit length between bottom and middle wire

$F_2 = \frac{\mu_0}{4\pi}\frac{2\left(\frac{1}{4}\right)\left(\frac{1}{5}\right)k^2}{r_2}$. As the forces are equal and opposite

so $F_1 = F_2 \Rightarrow \frac{r_1}{r_2} = \frac{5}{3}$

14)



B_1 = magnetic field int. due to AOB = $\frac{\mu_0 I_1}{2\pi d}$, due to COD $B_2 = \frac{\mu_0 I_2}{2\pi d}$

$B_{Net} = \sqrt{B_1^2 + B_2^2} = \frac{\mu_0}{2\pi d}\sqrt{I_1^2 + I_2^2}$

15) $B_{axis} = \frac{\mu_0 IR^2}{2[R^2 + x^2]^{3/2}}$

$= \frac{4\pi \times 10^{-7} \times 2.5 \times (3 \times 10^{-2})^2}{2[(3 \times 10^{-2})^2 + (4 \times 10^{-2})^2]^{3/2}}$

$\Rightarrow B_{axis} = 36\pi \times 10^{-7}\hat{k}$

16)

Heat supplied by steam = mL ...(1)
Heat absorbed by water = 20 × 1 × (100 – 46)
= 20 × 54 ...(2)
L = 540 cal/g
by equation (1) & (2)
m × 540 = 20 × 54
m = 2
So total mass of water is = 2 + 20 = 22 g

17) $r = \frac{mV\sin 30^\circ}{qB}$
$= \frac{3 \times 10^5}{10^8 \times 0.3} \times \frac{1}{2}$
$= 0.5 \times 10^{-2}$ m
r = 0.5 cm

18) The efective magnetic field at O
$B = B_{PE} + B_{RS} = \frac{\mu_0}{4\pi} \cdot \frac{3}{2}\frac{\pi I}{R} + \frac{\mu_0}{4\pi} \cdot \frac{\pi}{2} \cdot \frac{I}{2R}$
$\Rightarrow \quad B = \frac{\mu_0 I}{4R}\left[\frac{3}{2} + \frac{1}{4}\right] = \frac{7}{16}\frac{\mu_0 I}{R}$
As per Fleming's Right Hand rule, direction of magnetic filed is perpendicular and in the plane of paper.

19) $e = Bv\ell = 0.1\ (2 \times 10^{-2}) \times 6$
$= 12 \times 10^{-3} = 12$mV

20)

$e = Bv\ell$
= (1)(1)(2)
⇒ e = 2 Volt

21)

$I = I^0\left(1 - e^{-t/\tau}\right)$ and $\tau = \frac{L}{R},\ I_0 = \frac{E}{R}$

22)

$V_L = V_C$
$\Rightarrow X_L = X_C$ (Resonance condition)
$i = \frac{V}{R} = \frac{100}{50} = 2A$
$V_R = iR = 2 \times 50 = 100$ V

23)

$X_L = \omega L = 0.4 \times 2\pi \times \frac{30}{\pi} = 24\Omega$

$Z = \sqrt{10^2 + 24^2} = 26\Omega$

$P_{avg} = V_{rms} I_{rm} \cos\phi = V_{rms}\left(\frac{V_{rms}}{Z}\right)\left(\frac{R}{Z}\right)$

$= \frac{RV_{rms}^2}{Z^2} = \frac{5}{8}W$

24)

Displacement current arises when electric field in a region is changing with time, which is given by $I_D = \varepsilon_0 \frac{d\phi_E}{dt}$. It will be so if the charge on a capacitor is not constant but changing with time.

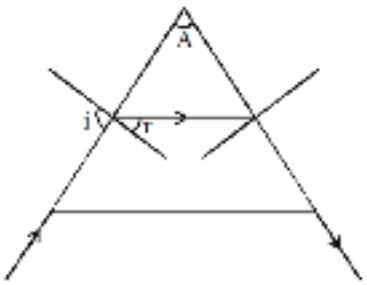
25) According to Bohr atomic radius

$r_n \propto n^2$

$\frac{r_2}{r_1} = \left(\frac{n_2}{n_1}\right)^2 = \left(\frac{2}{1}\right)^2 = 4$

$r_2 = 4r$

26)



When the angle of incidence and emergence are equal, light ray travel parallel to base inside prism.

i = 90°, r = 90 – A

for equilateral prism

A = 60°

r = 30°

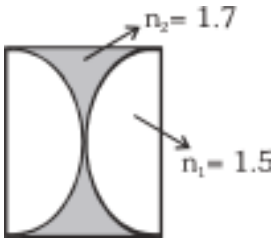
Snell's law at surface 1

1·sin 90° = μ·sin30°

μ = 2

27) $\frac{1}{V} + \frac{1}{U} = \frac{1}{f}$

Valid for paraxial rays.

28) 

From lens maker's formula

$$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

we have $\frac{1}{f_1} = (1.5 - 1)\left(\frac{1}{20}\right) = \frac{1}{40}$

$$\frac{1}{f_2} = (1.5 - 1)\left(\frac{1}{20}\right) = \frac{1}{40}$$

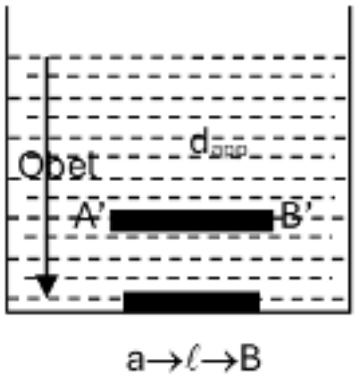
& $\frac{1}{f_3} = (1.7 - 1)\left(\frac{2}{-20}\right) = -\frac{7}{100}$

Now $\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3}$

$\Rightarrow \frac{1}{f} = \frac{1}{40} + \frac{1}{40} - \frac{7}{100}$

$\Rightarrow$ f = – 50 cm

29)



there is no change in horizontal length of rod. So, its remains same.

30) After 10 sec.

u = –80 cm & f = –100 cm

$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow$ v = 400 cm

31) $\frac{1}{2f_2} = \frac{1}{f_1} = (\mu_1 - 1)\left(\frac{1}{\infty} - \frac{1}{-R}\right)$

$$\frac{1}{f_2} = (\mu_2 - 1)\left(\frac{1}{-R} - \frac{1}{\infty}\right)$$

$\frac{(\mu_1 - 1)}{R} = \frac{(\mu_2 - 1)}{2R} \Rightarrow 2\mu_1 - \mu_2 = 1$

32) (A) When the source (s) moves up, the central fringe move downward at a point at which path difference is zero. Moving the source does not change wavelength (λ), the distance to the screen (D) and slit separation (d). So, fringe width remains constant A → [Q, R]
(B) A thin glass slab is placed in front of slit S_2, it will increases the optical path length of light passing through it. The delay requires the central fringe to move downward toward S_2 to equalize the path. Introducing a thin glass slab shift the entire pattern but does not change the spacing between fringes (β). So, it remains constant B → [Q, R]
(C) Whole apparatus is immersed in water then

$$\beta' = \frac{\beta}{\mu} \quad \left(\because \lambda_{water} = \frac{\lambda_{air}}{\mu}\right)$$

∴ C → [S]
(D) Diameter of slit S_2 is doubled intensity of light from slit is proportional to its area (or width)
So, intensity increases.
Fringe width remains constant D → [R, T]

33) • Minima Condition (Destructive Interference): $a \sin\theta = n\lambda$
where n is the order of the minimum (n = 1,2,3....).
The first minimum for wavelength λ_1 occurs at:

$$a \sin\theta = 1 \cdot \lambda_1 = \lambda_1 \quad .....(1)$$

• Maxima Condition (Secondary Maxima, Approximate Formula):

$$a \sin\theta = \left(n + \frac{1}{2}\right)\lambda$$

For the second maximum (n = 2) of wavelength λ_2:

$$a \sin\theta = \left(2 + \frac{1}{2}\right)\lambda_2 = 2.5\lambda_2 \quad .....(2)$$

Since both diffraction conditions coincide at the same angle θ, we equate the two expression:

$$\lambda_1 = 2.5\lambda_2$$

From Eq. (1) and (2)

34)

$I = I_0\cos^2\phi$, $\phi = 60°$

$$\xrightarrow{(I_0)} P_1 \xrightarrow{\left(\frac{I_0}{2}\right)} P_2 \xrightarrow{\left(\frac{I_0}{8}\right)} P_3 \xrightarrow{\left(\frac{I_0}{32}\right)} P_4 \xrightarrow{\left(\frac{I_0}{128}\right)} P_5 \xrightarrow{\left(\frac{I_0}{512}\right)}$$

35)

$X^{220} \longrightarrow Y^{216} + \alpha^4$

$K_y + K_\alpha = Q = 12.25$ MeV

$$\frac{K_y}{K_\alpha} = \frac{m_\alpha}{m_y} = \frac{4}{216}$$

∴ $K_\alpha = \frac{216}{220} \times 12.25$ MeV

$K_\alpha = 12.03$ MeV

36)

A particle which has more binding energy per nucleon is more stable.
Here α-particle has more binding energy per nucleon than deuctron.
So, a-particle is more stable.

37)

For λ_1 = 2475 Å

$E_1 = \frac{12400}{2475}$ = 5.01 eV

For λ_2 = 6000 Å

$E_2 = \frac{12400}{6000}$ = 2.06 eV

Only λ_1 will emit electron

KE = $E_1 - \phi$

KE = 5.01 – 4.8 = 0.21 eV

$r = \frac{\sqrt{2mK}}{QB}$

$r = \frac{\sqrt{2 \times 9.1 \times 10^{-31} \times 0.21 \times 1.6 \times 10^{-19}}}{1.6 \times 10^{-19} \times 3 \times 10^{-5}}$

r = 5.1 cm

38) $\lambda = \frac{h}{\sqrt{2mE}}$

39)

$\lambda_e = \lambda_{ph}$

$\frac{h}{P_e} = \frac{h}{P_{ph}}$

$P_e = P_{ph}$

$m_eV = \frac{E_{Ph}}{C}$

$E_{Ph} = m_eVC$ .....(1)

$(kE)_e = \frac{1}{2}m_eV^2$ .....(2)

From (1) and (2)

$\frac{(KE)_e}{E_{Ph}} = \frac{\frac{1}{2}m_eV^2}{m_eVC} = \frac{\frac{1}{2}.\frac{C}{4}}{C} = \frac{1}{8}$

40)

After doping impurity atom behave as a immobile ion so total charge is zero.

41) $I_Z = \frac{P_Z}{V_Z} = \frac{600 \times 10^{-3}}{24} = 25 \times 10^{-3} A$

Voltage drop across R = 32 – 24 = 8V

$R = \frac{8}{25 \times 10^{-3}} = 320\,\Omega$

42) (i) $I = \frac{V - V_0}{R_{net}} = \frac{20 - 0.7}{510}$ = 37.8 mA

(ii) $V = 20 - V_0 - IR$

$= 20 - 0.7 - (37.8 \times 10^{-3}) \times 10$

$= 20 - 0.7 - 0.34 = 18.9$ V

When diode is ideal then $V_0 = 0$, R = 0

43)

V = iR

$3 = 3 \times R \Rightarrow R = 1\Omega$

44) $i_1 = \frac{E}{R + G}$

$i_2 = \frac{E}{R + \frac{GS}{G+S}} = \frac{E(G+S)}{RG + RS + GS}$

$\frac{i_1}{2} G = \left(i_2 - \frac{i_1}{2} \right) S$

$\frac{i_1}{2}(G + S) = \frac{E(G+S)}{RG + RS + GS}.S$

$\frac{1}{2(R+G)} = \frac{S}{RG + RS + GS}$

2RS + 2SG = RG + RS + GS

RS + GS = RG

S(R + G) = RG

45)

Real depth = (23.25 – 5.25) cm

= 18.00 cm Apparent depth

= (23.25 – 11.25) cm = 12.00 cm

$RI = \frac{\text{Real depth}}{\text{App. depth}} = \frac{18.00}{12.00} = 1.5$

CHEMISTRY

46)

Brown ring complex

$[\overset{+1}{Fe}(H_2O)_5 \overset{+1}{NO}]SO_4$

47)

chromyl chloride test

48)

$Cu^{+2}+K_4[Fe(CN)_6] \rightarrow Cu_2[Fe(CN)_6]$ Chocolate brown ppt

$Cu^{+2} + NH_3$ excess $\rightarrow [Cu(NH_3)_4(H_2O)_2] (OH)_2$

Switzer salt

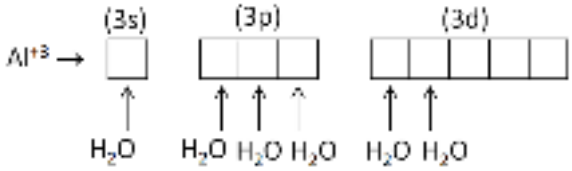
$Cu^{+2} + NH_4OH \rightarrow Cu(OH)_2$ (Blue ppt)

49)

• Assertion "Test of (NH_4^+) can not be done within group analysis=" → True.
Reason: (NH_4^+) is tested before group analysis because ammonium salts are added during group separation steps which interferes with its detection.

50) $[AI(H_2O)_6]^{+3}$

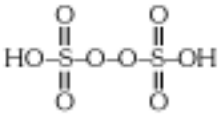
$Al^{+3} \rightarrow$ [Ne] $3s^o$ $3p^o$



Hybridisation is sp^3d^2

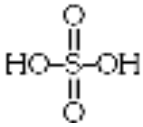
51)

A → Peroxodisulphuric acid

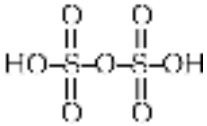


B → Sulphuric acid

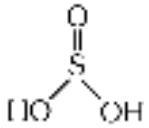
H_2SO_4

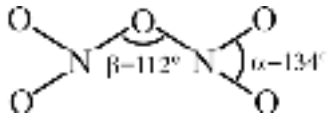


C → Pyrosulphuric acid $H_2S_2O_7$



D → Sulphurous acid H_2SO_3



52) 

53) $[Cr(NO_3)_3(NH_3)_3]$ has two stereoisomers fac and mer-isomers. Both are optically inactive

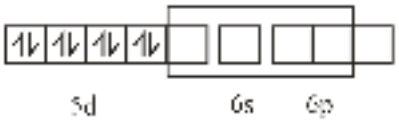
$K_3[Fe(C_2O_4)_3]$ has two stereoisomers. Both are optically active One is d from and the other is 1 form

$[CoBrCl(en)_2]$ has three stereoisomers. Two are geometrical isomers cis and trans.

The cis isomer has two optical isomers 'd' from and 'l' from
Trans isomer is meso in nature
The compound $[CoCl_2(Ox)_2]^{3-}$ has three stereoisomers. Two are geometrical isomers cis and trans.
The cis isomer has two optical isomers 'd' from and 'l' from
Trans isomer is meso in nature

54)

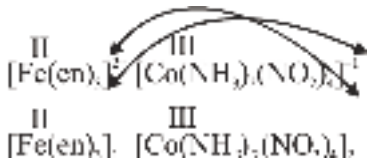
$[PtCl_4]^{2-}$ involves dsp^2 hybridization with orbitals of different n values.
$[PtCl_4]^{2-}$

$Pt^{+2} = 5d^8 =$ 

55) **Explanation:**Name complex anion with multiple ligands including cyanide.
Concept: Cation first, then anion, ligands alphabetical, indicate mate oxidation.
Solution:

A. Complex: $K_2[Cr(CN)_2O_2(O)_2(NH_3)$ → anion: $[Cr(CN)_2O_2(O)_2(NH_3)^{2-}$
B. Ligands:
C. NH_3 → ammine
D. CN^- → cyano
E. O_2^{2-} → peroxo
F. O^{2-} → oxo
G. Alphabetical order: ammine, cyano, oxo, peroxo
H. Metal: Cr in anion → suffix –ate, oxidation state +VI
I. Full name: Potassiumamminedicyanodioxoperoxochromate(VI)

Final Answer: Option (1)

56) 

57)

C-C bond length
C_2H_4 < $K[PtCl_3(C_2H_4)]$

58)

Ease of liquification
He < Ne < Ar < Kr < Xe

59)

$E^0_{M+2/M}$ = V(–1.18), Mn(–1.18), Fe(–0.44),
Co(–0.28), Ni (–0.25)

60)

Properties of d-block.

61)

$r = k[A]\,[B]^{0.5}$
Order of reaction = 1 + 0.5
$= \frac{3}{2}$
Unit of $K = \left(\frac{mol}{L}\right)^{1-n} sec^{-1}$
$= \left(\frac{mol}{L}\right)^{1-\frac{3}{2}} sec^{-1}$
$= \left(\frac{mol}{L}\right)^{-\frac{1}{2}} sec^{-1}$
$mol^{-1/2}\ L^{+1/2}\ Sec^{-1}$

62)

• Amount Remaining after 'n' half-lives: Amount Remaining= Initial Amount × $\left(\frac{1}{2}\right)^n$

• Substitute the values: Amount Remaining = 10 gm × $\left(\frac{1}{2}\right)^6$ Amount Remaining = 10 g ×

$\left(\frac{1}{64}\right)$ Amount Remaining = 0.15625 gm
used amount (10 – 0.156) ⇒ 9.844 gm

63)

$2H_2O \rightarrow 2H_2O + O_2$
$(ROD)H_2O_2 = 0.68 \text{ kg hr}^{-1}$
$= \frac{0.68 \times 10^3}{34} \text{ mol hr}^{-1}$
$= 20 \text{ mol}^{-1} \text{ hr}^{-1}$
$(ROA)H_2O = (ROD)H_2O_2$
$= 20 \text{ mol}^{-1} \text{ hr}^{-1}$
$= \frac{20 \times 18}{10^3} \text{kg hr}^{-1}$
$= 0.36 \text{ kg hr}^{-1}$.
$(ROA)_{O_2} = \frac{(ROA)H_2O}{2}$
$= 10 \text{ mol hr}^{-1}$.
$= \frac{16 \times 32}{10^3} \text{kg hr}^{-1}$
$= 0.32 \text{ kg hr}^{-1}$.

64) $\frac{P_A^o}{P_B^o} = \frac{1}{2}, \frac{X_A}{X_B} = \frac{1}{2}$
$P_S = P_A^o X_A + P_B^o X_B$
$P_A^o X_A = Y_A P_S$
$\frac{1}{Y_A} = \frac{P_S}{P_A^o X_A}$
$\frac{1}{Y_A} = \frac{P_A^o X_A + P_B^o X_B}{P_A^o X_A}$
$= 1 + \left(\frac{P_B^o}{P_A^o}\right)\left(\frac{X_B}{X_A}\right)$
$\frac{1}{Y_A} = 1 + \frac{2}{1} \times \frac{2}{1} = 1 + 4 = 5$
$\Rightarrow Y_A = \frac{1}{5} = 0.2$

65) According to Raoult's law, adding a non-volatile solute lowers the vapour pressure of a solvent.
The higher the concentration of the solute, the lower the vapour pressure of the resulting solution at a given temperature.
From the graph, at any fixed temperature, Curve III has the lowest vapour pressure, followed by Curve II, then Curve I.
Since Curve III shows the maximum lowering of vapour pressure, it must have the highest concentration.

Therefore, the concentration order is: III > II > I.

66) **Leclanche cell (a):** A common dry cell that uses an electrolyte of ammonium chloride and zinc chloride (q).
Mercury cell (b): A small primary cell with an electrolyte paste of potassium hydroxide and zinc oxide (p).
Lead Storage battery (c): A rechargeable battery that uses a 38% solution of sulfuric acid (s).
Ni-Cd Cell (d): A rechargeable battery that uses a moist electrolyte of potassium hydroxide (r).

67)

(P) Aqueous NaCl with Pt electrode → Anode: Cl_2, Cathode: H_2 → (iv)
(Q) Dilute H_2SO_4 with Pt electrode → Anode: O_2, Cathode: H_2 → (iii)
(R) Aqueous NaCl with Hg electrode → Anode: Cl_2, Cathode: Sodium amalgam → (i)
(S) Molten NaCl → Anode: Cl_2, Cathode: Na → (ii)

68)

(A) $2H_2O \rightarrow O_2 + 4H^+ + 4e^-$
 1 mole of O_2 required 4F of charge.
 At STP 1 mole (22.4L) O_2 required 4F of charge.
 For 5.0 L O_2 moles of O_2 $= \frac{5.6}{22.4} = 0.25$ mol
 Charge required = 0.25 × 4 = 1F (R)

(B) $Cr_2O_7^{-2} + 14H^+ + 6e^- \rightarrow 2Cr^{+3} + 7H_2P$
 1 mole of $Cr_2O_7^{-2}$ requires 6 F.
 For 0.5 mole $Cr_2O_7^{-2}$ = 0.5× 6 = 3F (P)

(C) The Reduction of Al^{+3} to Al requires 3 moles of electrons per mole of Al.
 For 18 g Al (0.6667 moles) 0.6667×3 = 2 F charge
– charge required = 2 F (S)

(D) 1 mole of Cu^{+2} requires 2 moles of electrons to be deposited as Cu.
– In 2L of 1 M $CuSO_4$, there are 2 × 1 = 2
moles of Cu^{+2}.
For 2 moles of Cu^{+2} :- 2 × 2 = 4F charges.
Charge required = 4F (Q)

69)

(i) → Correct
(ii) → False as $Fe(OH)_3$ is a positive sol.
(iii) → Correct
(iv) → False as lyophobic sols are irreversible.

Statement (i) and (iii) are correct.

70)

Equivalent weight = Molar mass / n-factor (electrons exchanged)
$Fe^{2+} \rightarrow Fe^{3+}$; change in oxidation state = 1 → n-factor = 1;
Equivalent weight = 392 / 1 = 392

$$FeSO_4.(NH_4)_2SO_4.6H_2O + KMnO_4 \xrightarrow{H^+} Mn^{+2} + Fe^{+3}$$

71)

1. Given SOPs :
   X = –0.65 V
   Y = –1.40 V
   Z = +0.36 V
2. Reducing power (of metals) : A higher (more positive) SOP means a stronger reducing agent.
   Order of SOPs: Z(+ 0.36 V) > X (– 0.65 V) > Y (– 1.40 V)
   Order of reducing power : Z > X > Y.
   $E°_{red}(X)$ = + 0.65 V
   $E°_{red}(Y)$ = + 1.40 V
   $E°_{red}(Z)$ = – 0.36 V
  Order of Oxidising power : $Y^+ > X^+ > Z^+$.
  Z has max SOP thus it can reduce both X^+ and Y^+.

Final answer : option (3)

72) Use $\Delta G° = -nFE°$ additivity for sequential reactions.

$$E^o_{total} = \frac{n_1E^o_1 + n_2E^o_2}{n_{total}}$$

$$E^0 = \frac{0.15 + 0.5}{2} = \frac{0.65}{2} = 0.325\,V.$$

73)

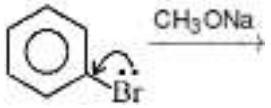
Cell: Ni(s) | Ni^{2+}(0.16 M) || Ag^+(0.002 M) | Ag(s)
$E°_{cell}$ = 1.05 V, n = 2
$Q = [Ni^{+2}]/[Ag^+]^2 = 0.16 / (0.002)^2 = 40000$
$\log Q \approx \log(4\times10^4) = 0.602 + 4 = 4.602$
$E_{cell} = 1.05 - (0.0591/2) \times 4.602 \approx 1.05 - 0.127 \approx 0.923$ V

74)

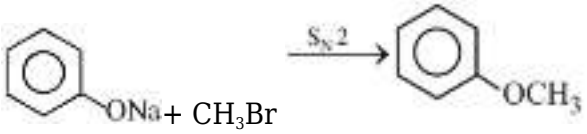
$\Delta T_f = i\, k_f.m$
$i = 1 + (n - 1)\,\alpha$
$= 1 + (2 - 1)\,0.2 = 1.2$
$\Rightarrow \Delta T_f = 1.2 \times 1.86 \times 0.2$
$= 0.4464$

$\approx 0.45°C$
So $T_f = -0.45°C$

75) $\pi_1 = \pi_2$
$i_1c_1 = i_2c_2$
$i_1 \times \frac{17.4}{174} = 2 \times \frac{5.85}{58.5}$
$i_1 = 2$
$1 + (3 - 1)\alpha = 2$
$\alpha = \frac{1}{2} = 50\%$

76) (x) 

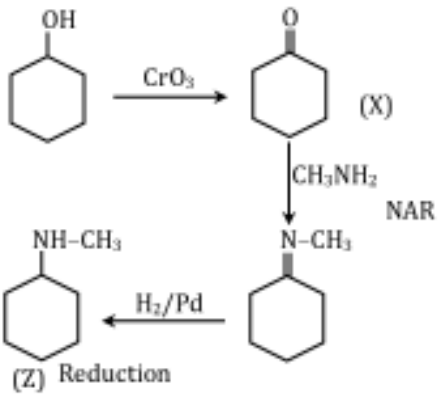
No reaction
Due to resonance C–Br bond acquires double bond character

(y) 

+ CH_3Br

Final answer: option (2)

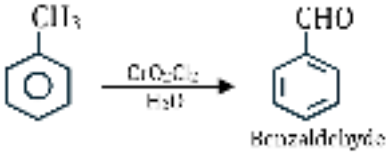
77) 1° and 2° alcohols are oxidised to corresponding acids by oxidation with potassium dichromate and its orange color turns green.
Oxidation of 3° alcohol does not take place. So it does not give a dichromate test.
Thus, (R) correctly explains (A).
Final Answer: Option (3) (tertiary alcohol)

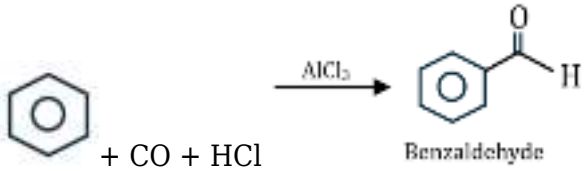
78) 

Final Answer: Option (4)

79)

(i) $R-C\equiv N + SnCl_2 + HCl \longrightarrow R-CH=NH \xrightarrow{H_3O^+} RCH=O + NH_3$

(ii) 

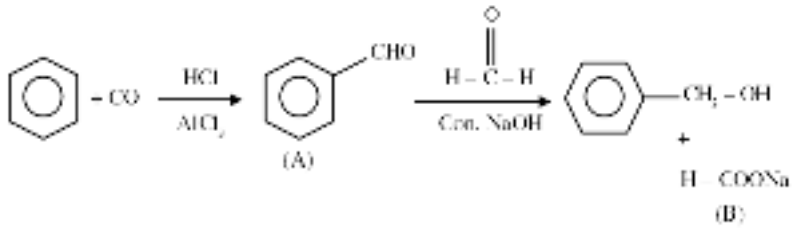
(iii) 

+ CO + HCl

(iv) $Ph-CH=N-CH_3 \xrightarrow{H_3O^+} Ph-CHO + CH_3-NH_2$

Final Answer : Option (4)

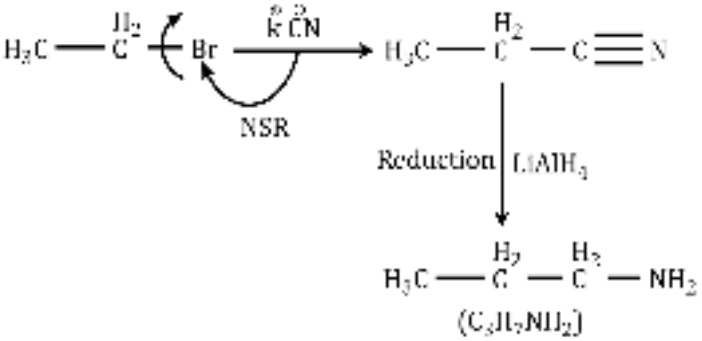
80)



Gattermann koch reaction proceed through ESR mechanism In Cannizzaro reaction oxidation of HCHO and reduction of benzaldehyde takes place.

Final Answer :- 2

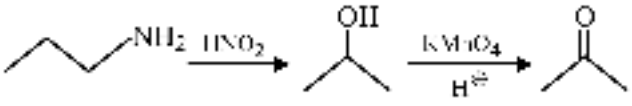
81) $C_2H_5Br \xrightarrow{X} product \xrightarrow{Y} C_2H_7NH_2$



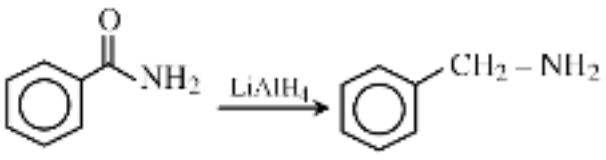
So x ⇒ kCN

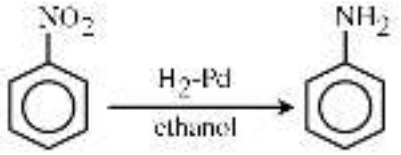
y ⇒ $LiAlH_4$

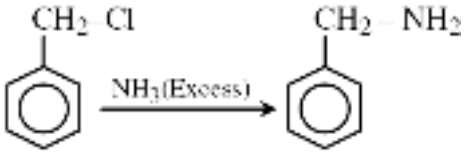
Final Answer - Option 1

82) 

83)

(1) 

(2) 

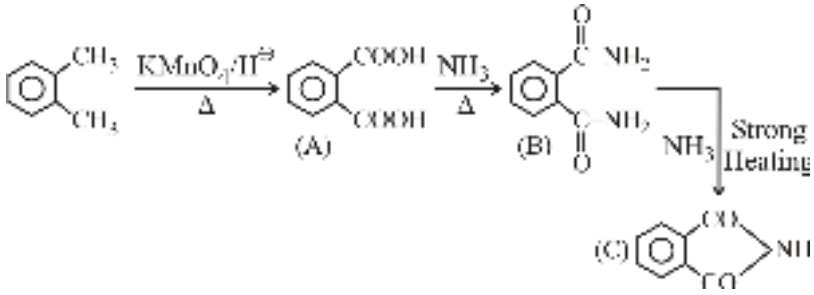
(3) 

(4) $CH_3-CH_2-C\equiv N \xrightarrow[(2)H_2O]{(1)DIBAL-H} CH_3-CH_2-CH=O$

Final Answer :- (4)

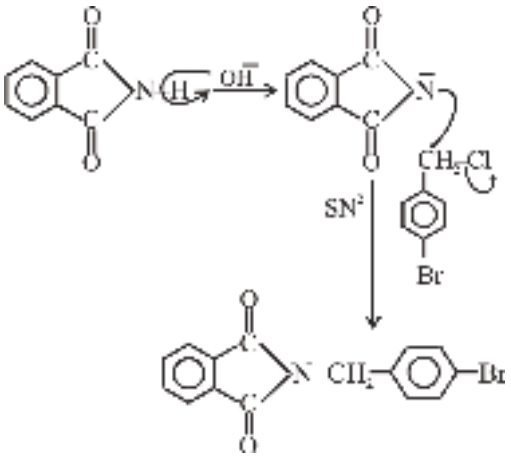
CH_3–CH_2–C≡N $\xrightarrow[(ii)\,H_2O]{(i)\,DIBAL-H}$

84) Solution : Gabriel phthalimide synthesis is used to preparation of aliphatic primary amine.
Final Answer : Option (3)

85) 

86)

Solution :- This is Gabriel pthalimide synthesis.



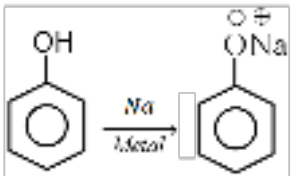
Final Answer :- Option (1)

87) NCERT XII II part Pg.# 355

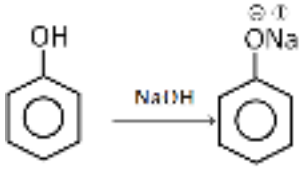
88) Primary alcohol = Red
Secondary alcohol = Blue

Tertiary alcohol = Colourless

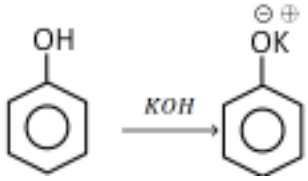
89)

1 . 

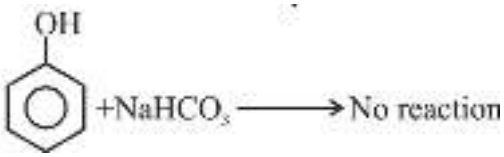
(strong acid)

2. 

+ H_2O (weak acid)
(strong acid)

3. 

+ H_2O (weak acid)
(strong acid)

4. 

because phenol is weaker acid then H_2CO_3
Final answer: option (4)

90)

On moving top to bottom basic nature decrease but nucleophilic nature increase.

BIOLOGY

91)

NCERT XI Pg.# 61

92)

NCERT XI Pg.# 71, 72

93)

NCERT XI Pg.# 75

94)

NCERT XI Pg.# 65, 66

95)

NCERT XI Pg.# 62

96)

NCERT XI Pg.# 76-77

97)

It is a diagram of shoot apical meristem.

98)

NCERT XI Pg.# 75-76

99)

NCERT-XII Pg.# 19

100)

NCERT-XII Pg.# 18-19

101)

NCERT-XII Pg.# 19-20

102)

NCERT-XII Pg.# 10

103)

NCERT-XII Pg.# 11

104)

NCERT-XII Pg.# 12

105)

NCERT-XII Pg.# 12

106)

NCERT-XII Pg.# 74

107)

NCERT-XII Pg.# 57-58

108)

NCERT-XII Pg.# 76

109)

NCERT-XII Pg.# 74

110)

NCERT-XII Pg.# 67

111)

NCERT-XII Pg.# 70

112)

NCERT-XII Pg.# 72

113)

NCERT-XII Pg.# 81, 82

114)

NCERT-XII Pg.# 80-81

115)

NCERT-XI Pg.# 107

116)

NCERT-XII Pg.# 93

117)

NCERT-XII Pg.# 100-101

118)

NCERT-XII Pg.# 91

119)

NCERT-XII Pg.# 94-95

120)

NCERT-XII Pg.# 97

121)

NCERT-XII Pg.# 210

122)

NCERT-XII Pg.# 212-213

123)

NCERT-XII Pg.# 207-208

124)

NCERT-XII Pg.# 207-208

125)

NCERT-XII Pg.# 219

126)

NCERT-XII Pg.# 199

127)

NCERT-XII Pg.# 194

128) NCERT-XII Pg.# 197

129)

NCERT-XII Pg.# 201

130)

NCERT-XII Pg.# 192

131)

NCERT-XII Pg.# 222

132)

NCERT-XII Pg.# 222

133)

NCERT-XII Pg.# 222-223

134)

NCERT-XII Pg.# 219

135)

(a) Members of pioneer community has less nutrient demand.
(d) Climax community is always predictable.
So a and d are false.

136) NCERT Page No. 172

137) NCERT XII, Pg. # 179

138)

NCERT Pg # 183

139) NCERT XII Pg. No. # 182

140) NCERT-XII, Pg # 213, 214 (E), 234, 235 (E)

141) NCERT Pg. # 174

142) NCERT, Pg. # 165, 166

143)

NCERT-XII, Pg. # 170

144)

NCERT Pg # 199

145)

NCERT-XII, Pg. # 169

146)

NCERT XII, Pg # 208

147)

NCERT-XII, Pg. # 180

148)

NCERT-XII, Pg. # 185

149) NCERT-XII, Pg. # 157

150) NCERT-XII, Pg. # 27, 28

151) NCERT-XII, Pg. # 35

152)

Module

153) NCERT Pg.# 28, 37

154) NCERT Pg.# 30

155) NCERT Pg.# 33

156) NCERT Pg. # 28-29

157) NCERT XII Page No.# 31, 33

158) NCERT Pg. # 35-36

159) NCERT-XII, Pg # 36

160) NCERT, Pg. # 46, 47

161) NCERT, Pg. # 43, 44, 45

162)

NCERT, Pg. # 41, 42

163) NCERT Pg.#44

164) NCERT page No. - 48

165) NCERT Pg. # 120

166)

NCERT Pg. # 118

167) NCERT Pg.# 113-116

168) NCERT Pg. # 124

169) NCERT ; Page No. # 112

170) NCERT Pg. # 123-124

171) NCERT Pg.# 140

172) NCERT Pg. # 133

173) NCERT, Pg. # 143

174) NCERT Pg. # 138

175) NCERT Pg. # 136

176) NCERT, Pg. # 141-145

177) NCERT, Pg. # 141

178) NCERT, Pg. # 131

179) NCERT Pg. # 137

180)

NCERT Pg. # 141